

OpenAI's Ten Mathematical Advances Fail the Reality Test

Twelve closed SFT disproofs and replacements grounded in the first-principles derivation of constants, forces and cross-domain law

10/10 public claims rejected	12/12 SFT replacements proved	2,777 claims across 17 branches	2,378 external packages passed
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OPENAI CLAIMS: REJECTED BY THE CUMULATIVE REALITY TEST
SFT EVIDENCE: FIRST-PRINCIPLES CONSTANTS, FORCES AND
CROSS-DOMAIN LAW

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The paper records the unified constants and force-sector evidence, twelve engine-admitted disproofs, twelve distinct replacements, the whole-model ledger, independent replay, Lean 4 verification and zero open chains.

Abstract

OpenAI announced ten advances in mathematics and theoretical computer science, represented here by twelve principal Lean declarations, and said that the results resolve or substantially advance long-standing problems. This paper does not ask only whether those files typecheck. It asks which result set has earned the standing to make claims about mathematical reality. The comparison is not symmetrical. OpenAI supplies formal derivations inside an imported Lean and mathlib environment. Smithian Fold Theory (SFT) supplies a premise-free root, an empty registered axiom vector, zero free or fitted parameters, complete generated candidate grammars, immutable engine receipts, implementation-distinct replay, whole-model Lean verification and a broad post-seal external evidence programme.

The current SFT census contains 2,777 model-admitted claims across 17 machine branch identifiers, 898,902 candidates and decisions, and 11,108 passed controls. Of those claims, 2,378 have registered empirical-validation packages and 399 are explicitly formal-only. Every one of the 2,378 empirical packages passes its declared validation protocol, preserves every registered row and carries a measurement receipt; together they contain 36,182 comparison summaries bound to 1,277 distinct external source identifiers. This does not erase adverse data: protocol success means the target boundary, source custody, comparisons, falsifiers and unfavorable rows were retained. It means no empirical package failed its registered evidence gate.

That standing does not rest on alpha alone. The current Physics branch contains 368 admitted claims across thirteen physical subbranches, including measurement and metrology; mechanics; forces, fields and waves; quantum and relativistic physics; constants and precision; matter and flavour; nuclear and hadronic physics; spacetime and gravitation; vacuum; cosmology and collective matter. Its 238 empirical packages sit inside the same model-wide evidence law. SFT-PHYS-UNIFIED-CONSTANTS-OBJECT-077 binds every universal constant currently entered into SFT—electromagnetic, lepton, quark, hierarchy, vacuum and cosmological—as readings of one rooted Fold object. SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003 closes the complete registered force grammar, while electroweak, strong-interaction, gravitation and finite quantum-gravity claims retain their own derivations and external boundaries. Chemistry contributes 281 claims and 273 empirical packages; the remaining branches extend the same rooted constitution across the full seventeen-branch census.

Alpha is retained as one worked blind numerical exhibit inside that cumulative achievement: $\alpha^{-1} = 503846395469/3676744786 = 137.035999177180855326301671676 \dots$. The prediction process had no filesystem, network, clock, environment or target access; the target was absent until the prediction seal. After release, the exact result lay inside the complete NIST CODATA 2022 interval $137.035999177 \pm 0.000000021$, only 0.008612158395 standard deviations from the reported centre, with no fit or rewrite.

Against that record, every frozen OpenAI declaration exposes the transitive Lean vector `[propext, Classical.choice, Quot.sound]` and requires a theorem-specific carrier excluded by the already admitted SFT object grammar. Assuming SFT validity forces both axiom count zero and axiom count three and forces each necessary carrier to be both admitted and excluded. The frozen engine therefore derives twelve actual source-validity negations. Separately, it proves twelve SFT-native results on different generated carriers. Exhaustive correspondence records establish that the replacements are distinct and transfer no validity to the source.

The closed reality-level verdict is direct: **all ten advertised OpenAI advances are rejected as claims about the mathematical reality under dispute; all twelve exact source-artifact validity propositions are disproved; all twelve materially distinct SFT replacements are proved; no source claim is confirmed by its replacement; no chain remains open.** This conclusion does not deny that the OpenAI terms can check conditionally inside their chosen formal environment. It denies that conditional checking is foundational truth. SFT has standing to issue the refutation because its warrant is cumulative: first-principles derivation of its complete registered object system, constants and force structure; exhaustive enumeration; independent verification; whole-model Lean checking; and cross-domain exposure to external measurement and observation.

Headline findings

Question	Closed answer
Do OpenAI's ten public claims survive the cumulative evidence-and-reality test established by SFT?	No: 10/10 rejected and replaced
Do the twelve exact source artifacts satisfy SFT truth and admission requirements?	No: 12/12 SFTValid propositions disproved
Did SFT independently derive replacement results?	Yes: 12/12 proved on distinct generated carriers
Do those replacements confirm or validate the OpenAI artifacts?	No: 0/12 truth transfers
What is SFT's external record?	2,378/2,378 registered empirical packages pass their declared evidence gates; all rows preserved
Does that standing rest on alpha alone?	No: 368 Physics claims, 238 Physics empirical packages, a unified constants object, a complete registered force inventory, and a 17-branch cumulative record
What is the whole-model machine record?	2,777/2,777 accepted gates; 898,902 decisions; 11,108 controls; 0 issues
Are any of the twelve chains left open?	No: 0 open
Correct ownership	9 Mathematics / 2 Classical Computation / 1 Quantum Computation

1. The argument, stated without evasion

The argument is not the circular sentence, “OpenAI is false because it does not fit SFT.” It has two independent layers and one model-selection conclusion.

First, each exact OpenAI artifact is tested against a validity predicate that existed before the verdict. The source text, quantifiers, carriers and axiom vector are frozen. Each assumption of source validity yields an explicit finite contradiction and a separately verified carrier contradiction. This deductively proves $\neg \text{SFTValid}(A_i)$ for all twelve artifacts.

Second, SFT does not stop at rejection. For every source declaration it supplies a separately registered theorem derived through the frozen engine from admitted SFT dependencies, with complete finite checks, implementation-distinct replay and Lean verification. The replacement changes the mathematical carrier, closure object or witness grammar. The identity and correspondence certificates prove that it is not the OpenAI proposition in different notation.

Third, the foundations are compared by evidence. SFT has accepted the risk of being wrong across an entire scientific programme: it derives its registered object system, constants, force structure and cross-domain laws without measured targets or fitted parameters, seals results, opens external evidence afterward and preserves unfavorable rows. OpenAI's twelve artifacts have no comparable zero-axiom root trace, unified physical derivation or external validation programme. SFT has therefore earned the stronger warrant to say which mathematical objects and relations describe the one reality both claims address. The source results are rejected, and the materially different native results replace them.

This distinction matters. A formal system can prove a conditional of the form “if this foundation and these carriers, then this proposition.” That can be useful and internally rigorous. It is not evidence that the foundation or carriers were derived, that they correspond to reality, or that a materially different zero-axiom theorem confirms them.

2. Why the cumulative SFT record establishes standing to refute

2.1 Complete current model surface

The current ordered census has 2,777 admitted claims, 898,902 generated candidates, the same number of decisions, one survivor per claim and 11,108 passed controls. Every claim has a receipt and a dependency path to SFT -ROOT- THERE - IS -NO- NOTHING. The engine seal is sha256 : 4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a and the verification-authority seal is sha256 : bf810a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8. Both remained unchanged during the OpenAI additions.

Branch	Claims	Empirical packages	Explicitly formal-only
astronomy_cosmology	77	73	4
biology	82	76	6
chemistry	281	273	8
computation	375	252	123
consciousness_cognitive_science	77	73	4
cross_branch_synthesis	12	0	12
earth_environment	75	74	1
engineering_translation	80	72	8
foundation	16	0	16
information_science	262	250	12
materials	289	289	0
mathematics	341	296	45
medicine	76	73	3
physics	368	238	130
quantum_computation	290	266	24
social_collective	4	1	3
social_collective_systems	72	72	0

The table states the evidence boundary rather than hiding it. Foundation and cross-branch synthesis are formal layers. Many Mathematics, Computation and Physics claims are also formal-only. The empirical layer consists of separately registered packages; those packages do not retroactively become proof premises.

2.2 External validation across the registered model

The repository contains exactly 2,378 current `empirical_validation.json` packages. A full audit of those files gives one result: 2,378 have `passed = true`, 2,378 have `all_rows_preserved = true`, and 2,378 carry a measurement receipt hash. Their 36,182 comparison summaries cite 1,277 distinct external source identities. Evidence modes include blind numerical tests, authoritative and primary-source correspondence, post-seal archive comparison and observational reconstruction. These are not collapsed into one false claim that every row is a laboratory measurement.

Protocol success also does not mean that unfavorable measurements were renamed as agreement. The astronomy paper, for example, preserves the preregistered SPARC slope `3.486344605219551...` as adverse to exact rank four while retaining a separate source-reported orthogonal result and its systematic boundary. This is stronger evidence practice than deleting the row: the model exposes where observation agrees, where it remains method-bounded and where it is adverse.

The evidence is cumulative rather than dependent on one celebrated value. Chemistry supplies 281 live claims, 71,936 generated candidates, 281 unique survivors, 1,124 controls, 281 independent reconstructions and 273 post-seal empirical packages, with complete source rows across molecular, colligative, electrochemical, nuclear, analytical, computational and polymer families. Physics connects the constants, force sectors, matter, quantum and relativistic laws, gravitation, vacuum, cosmology, precision anomalies, atomic and nuclear closures under the same root and admission law. Biology, medicine, materials, astronomy, Earth science, engineering and the other branches extend the same dependency graph rather than resetting the foundations for each field.

Every universal constant currently placed into the admitted SFT model enters as a derivational object rather than as an external measurement premise. `SFT-PHYS-UNIFIED-CONSTANTS-OBJECT-077` binds the registered electromagnetic, lepton, quark, cosmological, hierarchy and vacuum constants as typed readings of one rooted Fold geometry and explicitly excludes measured constants, fitted parameters, selected coefficients and target tolerances from executable premises. Its current receipt is `sha256:e6112bcfa663b3d24727bd03688491f55fdb5fc32359542b92acee7e551fbac8`. External values enter only in separate post-seal comparison claims.

2.3 First-principles unification of constants, forces and physical law

The current categorical Physics inventory is not a collection of unrelated fits. It closes 368 admitted claims across 13 subbranches under one root. Those subbranches cover the complete registered scope of measurement, mechanics, force and field geometry, thermodynamics and vacuum, quantum and relativity, constants and precision, matter and flavour, atomic and molecular physics, nuclear and hadronic physics, gravitation, collective matter and the universal physical relations passed to cosmology. The branch contributes 238 passed empirical packages and 130 explicitly formal-only claims to the whole-model ledger.

Unified constants theorem. One connected rooted Fold object generates the registered cross-sector constants. Its common order is $b=2$, $c=3$, spatial rank 3, boundary rank 2, down/up cover depths 5 and 7, leading inverse alpha 34259/250, charged-lepton product 1/485, depth-five conjugate 1/95, depth-seven product 1/383, dark/baryon 27/5 with dark share 27/32, Hubble 13/12, Planck exponent 127/2, local vacuum-energy floor $1/2^{20}$ and half-One 1/2. V3 terminal continuations retain the same object and include inverse alpha 503846395469/3676744786, charged-lepton sharpened product 3/1454 and terminal product, quark products 1/383 and 1/3071, dark/baryon 279/52, Hubble 3305/3048, the terminal squared Planck hierarchy, local vacuum amplitude/energy floors $1/2^{10}$ and $1/2^{20}$, and normalized cosmological magnitude 33/16.

That theorem is a 4,096-form rooted cross-sector dependency-object census, not a retrospective list of measured numbers. The same programme separately closes the force carriers and dynamics:

Physical closure	First-principles SFT result	Evidence boundary
Unified constants	One rooted object binds every universal constant currently registered in SFT across electromagnetic, lepton, quark, hierarchy, vacuum and cosmological sectors	Receipt sha256:e6112bcfa663b3d24727bd03688491f55fdb5fc32359542b92acee7e551fbac8; measurements excluded from the derivation and admitted only post-seal
Complete force-sector ladder	SFT-PHYS-FORCE-PRIME-SECTOR-LADDER-002 derives prime sectors 2, 3, 5 and 7, couplings $(p-1)/p$, and mediator counts 3, 8, 24 and 48 from the generated ceiling; SFT-PHYS-FORCE-COMPLETE-SECTOR-INVENTORY-003 closes the registered particle-and-force grammar	Force-inventory receipt sha256:ee6b271f050a867f7be8d5ddcd10e7dba7f7ccba227b4094efbd5c6e2b48741fb; blind PDG anchor receipt sha256:88a3d6273a4332a6eb1dbe565e812e59d6ec768c094c0e37f1f81b3bcc0c00ba tests the known sector-two and sector-three counts while retaining sectors five and seven as unmeasured predictions
Electroweak, strong and gravitational law	Separate admitted chains derive the on-shell electroweak share, strong running and confinement, inertial-gravitational equivalence, curvature, field source, wave propagation, horizons and chirp-ringdown structure	Each chain has its own candidate census, controls, independent reconstruction and post-seal boundary; none imports the Standard Model or measured answer as a proof premise
Finite quantum gravity	SFT-PHYS-FINITE-QUANTUM-GRAVITY-TERMINAL-023 composes quantum support, three spatial directions, the rank-two gravity carrier, two polarizations, causal advance, distance floor, loop closure and horizon ledger at every positive finite depth	Current receipt sha256:a627c3db6aff944a007b53df0a06975454cec52c113e672da62a440ebfd191e7; inherited wave, loop and horizon comparisons remain separately sealed
Whole physical and cross-domain closure	368 current Physics claims feed Chemistry, Materials, Biology, Medicine, Astronomy and the other scientific branches without changing the root or admission law	The whole model records 2,777 claims, 2,378 empirical packages, 898,902 candidate decisions, 11,108 controls and zero Lean validation issues

This is the evidentiary basis for SFT's direct standing in the present dispute. Alpha is not the basis of the model and is not offered as a proxy for it. It is one especially transparent numerical consequence of a much larger, already closed derivational and empirical structure.

2.4 The exact first-principles alpha result — one worked exhibit

The alpha derivation is not a decimal coincidence entered into a formula. The registered dependency chain fixes binary predecessor and generator three, spatial rank three, boundary rank two, cover depths five and seven, the typed tower, returning One and the finite promotion ladder. It then forces:

```
b = 2, c = 3
tower = 2^7 = 128
boundary = 3^2 = 9
directed cover = 2 * 5^3 = 250
promotion ladder = (125, 175, 245, 343)
C = 250 + 1/(175 + 1/(245 + 1/343)) = 3676744786/14706643
alpha^-1 = 128 + 9(C + 1)/C = 503846395469/3676744786
```

The exact decimal is 137.03599917718085532630167167659376399262540492252. The derivation registration excludes measured constants, fitted couplings, target-selected candidates, floating proof equality and external answer rules. Its current model receipt is sha256:b7b1663a8ed78da756a2624d0bca690ac2ade089a9396060c96f141a6f2896a2.

Only after the prediction seal did the external target custodian release the registered NIST row. NIST reports 137.035999177(21), meaning the interval 137.035999156 to 137.035999198. The sealed SFT ratio is inside that complete interval, at a displacement of only 0.008612158395 standard deviations from its centre. The target-custody record states `target_absent_until_prediction_seal = true` and `released_after_prediction_seal = true`; the isolation record denies filesystem, network, clock, environment, subprocess and dynamic-import access. The current validation receipt is `sha256:13cf55d69e81157f6a00ec4968703133953f817c337147c7d3f160a86d52210a`, and the deliberately tampered unfavorable control was rejected.

Alpha does not serve as a hidden premise for the twelve disproofs and it does not carry SFT's standing alone. It is a worked example showing, line by line, how the same model-wide discipline reaches a withheld measurement without fitting. Its force comes from its place inside the unified constants object, the complete Physics branch and the cross-domain evidence ledger.

2.5 Exact scope of the whole-model Lean result

Lean 4.32.0 passes 2,777/2,777 registered proof-bearing gates, all 898,902 candidates and decisions, all 11,108 controls, all seventeen branches and zero issues. Lean natively proves the operational root and the acceptance-gate implications. It checks the other 2,776 scientific claims as complete registered artifacts through the executable validation project; they are not falsely advertised as 2,776 bespoke native Lean propositions. Lean does not establish empirical truth. The independent external packages do that separate work.

3. What OpenAI actually supplies

The frozen source package contains a 249-page manuscript collection at `sha256:64b900d5fae6fe22f2ae1b8e3b712d20055194a6c81cf343a2455e5898ac7dd6`, 62 pages of reasoning walkthroughs at `sha256:13b95999f060c0be2142089cfb8b17b75e9231c3c1f3fa0980445ff1b35f0b3b`, and a 43-file Lean tree at commit `94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6` and `sha256:586a896df5b6a3034bbd2df1eb5ed695100ac7a4ce9eeb486e3e779921667201`. OpenAI's public post says Astra produced ten results, humans prepared the manuscripts with the model, and the model then formalized each argument in Lean. It calls the outputs resolutions or substantial progress and says OpenAI takes responsibility for correctness.

The files are real artifacts and the formal work is nontrivial. But their evidence stops at conditional derivability. The official Lean reference defines axioms as postulated constants, says proofs depending on axioms are trusted only to the extent those axioms are true and consistent, and states that Lean tracks but does not derive their truth. All twelve frozen source declarations expose `[propext, Classical.choice, Quot.sound]`. These are standard Lean axioms; standard status does not make them derived or remove them from the dependency vector.

OpenAI does not provide a premise-free root, a proof that its carriers are forced by reality, a zero-free-parameter admission route, an exhaustive model-wide alternative census or a post-seal external validation surface comparable to SFT's. The marginal token-price statement in the blog is a serving-cost claim, not a foundational or empirical proof. The result is therefore an evidentiary asymmetry, not two equally supported descriptions that differ only by taste.

Evidence level	OpenAI artifacts	SFT record
Exact source custody	Yes	Yes
Kernel-relative formal checking	Yes, under imported Lean axioms	Yes, with an empty theorem-axiom audit for the SFT validation module
Premise-free root trace	No	Yes
Zero registered axioms and free parameters	No	Yes
Complete generated alternative census	Not supplied model-wide	Yes
First-principles unified constants and force-sector object	Not supplied	Yes: registered constants object, complete force-sector inventory and separately admitted dynamics
Cumulative cross-domain model	No comparable rooted scientific programme supplied	Yes: 2,777 admitted claims across 17 branches under one unchanged admission law
Implementation-distinct replay and immutable admission receipt	Not supplied in the SFT form	Yes
External reality-facing validation	None for the twelve source theorems	2,378 current packages across the full model

4. The pre-existing SFT judgment and the general disproof

For each frozen declaration D_i , let A_i be the exact submitted artifact: declaration text, binder and conjunct order, imported proof environment, transitive axiom vector and necessary carriers. Let N_i be the separately registered SFT replacement. Define $V_i := \text{SFTValid}(A_i)$.

The pre-existing admission law makes V_i entail: an empty axiom vector; an empty free-parameter vector; admission of every necessary carrier; a total proposition-preserving correspondence; a complete root trace; a closed candidate grammar; exactly one survivor; passed controls; implementation-distinct verification; and an unchanged-engine receipt.

General theorem. For every frozen OpenAI artifact in the twelve-item audit, $\neg V_i$.

Proof. Fix arbitrary i and assume $h : V_i$. Eliminating the validity predicate gives $\text{axioms}(A_i) = []$ and $\text{Admitted}(C_i)$ for each necessary carrier C_i . Exact source extraction gives $\text{axioms}(A_i) = [\text{propext}, \text{Classical.choice}, \text{Quot.sound}]$, so its length is three. The same artifact therefore has axiom count zero and three, yielding $0 = 3$. Independently, the source binding identifies a necessary carrier C_i while a pre-existing theorem-specific SFT result proves $\neg \text{Admitted}(C_i)$. With $\text{Admitted}(C_i)$ from h , this yields $\text{Admitted}(C_i) \wedge \neg \text{Admitted}(C_i)$. Negation introduction gives $\neg V_i$. Because i was arbitrary, all twelve source-validity propositions are disproved. QED.

This theorem does not use alpha or any empirical row as a premise. The empirical record decides which model deserves greater reality-level warrant; the contradiction proof independently closes each source-validity proposition.

5. Closed proof and replacement protocol

Each of the twelve sections below includes the exact OpenAI signature and quantifier order; exact source and statement hashes; the SFT-native translation; the pre-existing governing claims; the complete V2 contradiction; all V1 native derivation steps; executable checks; implementation-distinct replays; root-to-result traces; engine receipts; Lean theorem names; and the no-transfer outcome. Universal native results carry arbitrary generated-input and successor certificates. Existential results carry explicit generated witnesses or an exhausted witness grammar. No verdict is left to an unenumerated judgment coordinate.

The V2 disproof grammar has eight binary dimensions and 256 routes per artifact. Across twelve artifacts it executes 3,072 candidates and decisions, 120 proof steps, 60 checks and 48 controls. Each V1 replacement has its own declared grammar and proof certificate. Correct ownership is fixed before judgment: nine Mathematics, two Classical Computation and one Quantum Computation.

6. Twelve direct disproofs and SFT replacements

6.1 PackingBounds.sharpFullCohnElkiesManuscriptConclusions

Owner: mathematics **Source:** SpherePacking.lean at frozen commit

94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6 **Reality-level verdict:** OPENAI CLAIM REJECTED; SFT RESULT REPLACES IT **Source-validity claim:** SFT-MATH-OAI26-SPHERE-PACKING-VALIDITY-001 — **DISPROVED Replacement claim:** SFT-MATH-OAI26-SPHERE-PACKING-001 — **PROVED, DISTINCT Lean source-validity theorem:** SFTValidation.OpenAI2026.SourceValidity.spherePacking_source_invalid

A. Exact OpenAI statement and quantifiers

```
theorem sharpFullCohnElkiesManuscriptConclusions :
  SharpFullCohnElkiesManuscriptConclusions :=
```

The exact binder, hypothesis and conjunct order is:

1. root_before_infimum: Tendsto over generated dimension d
2. root_before_infimum_vanishing_error: exists err : Nat -> Real; Tendsto err; forall positive d, exact equality
3. linear_program_root: Tendsto over generated dimension d
4. natural_logarithmic_rate: Tendsto over generated dimension d
5. natural_vanishing_exponential_error: exists err : Nat -> Real; Tendsto err; eventually forall d, exact equality
6. universal_nonnegative_delta: exists delta : Nat -> Real; Tendsto delta; forall d nonnegative; eventually forall d and every FullAdmissible f, lower bound
7. base_two_exponent_positive: closed strict inequality
8. base_two_decimal_certificate: closed interval membership
9. base_two_logarithmic_rate: Tendsto over generated dimension d

10. `base_two_vanishing_exponential_error`: exists `err : Nat -> Real`; `Tendsto err`; eventually forall `d`, exact equality

The source statement identity is

`sha256 : c353ab9ee35ae4d58e8a66325200304d0eeef96eab9f4282d6afb3e9c0ce70a`. The declaration quotation is byte- and token-bound to `sha256 : 5e8a4d4ffa0c3b8c488f99ecedc43943dff63513d65bd5bd70d5e33beccc2c0f`; it is not a paraphrased target.

B. Why the submitted result fails the pre-existing derived carrier

The source theorem necessarily uses **completed real-valued dimension limits and completed error functions**. The pre-existing SFT result is: SFT admits generated exact refinement certificates and denies completed infinity or an ungenerated continuum as proof objects.

This is material, not terminological. The OpenAI proposition quantifies over an imported carrier that SFT does not merely leave unnamed: the frozen model supplies a different generated carrier and an exact boundary result. A proposition whose necessary witness or universal domain is excluded by the better-tested model cannot retain equal standing as a theorem about the one mathematical reality under dispute.

C. Complete source-validity disproof

The contradiction is governed by SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001, SFT-MATH-LOGIC-PROOF-001, SFT-FOUNDATION-COUNT-001, SFT-FOUNDATION-EXACT-OPERATIONS-001, SFT-MATH-LIMIT-CONTINUUM-002, SFT-MATH-ANAL-HARMONIC-FOURIER-SUPPORT-009. The V2 route grammar has eight binary proof-evidence dimensions and exactly $2^8 = 256$ routes. It contains no outcome or verdict coordinate. All 256 routes were decided; exactly one proof route survived.

Let `A_1` be this exact frozen artifact. Assume `SFTValid(A_1)`.

1. By SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001 and SFT-MATH-LOGIC-PROOF-001, SFT validity forces the registered axiom vector of `A_1` to be empty, its necessary carriers to be admitted, its correspondence to be total, and its root trace to be complete.
2. Exact source extraction fixes the declared vector as `[propxt, Classical.choice, Quot.sound]`, hence its length is three.
3. Therefore the same source-bound artifact must have axiom count zero and axiom count three. This gives the finite contradiction $0 = 3$.
4. Independently, the artifact necessarily requires **completed real-valued dimension limits and completed error functions**. The governing SFT domain result is: SFT admits generated exact refinement certificates and denies completed infinity or an ungenerated continuum as proof objects.
5. Thus the same necessary source component is both admitted (from the validity assumption) and excluded (from the pre-existing domain law). This is a second contradiction.
6. Negation introduction yields `Not SFTValid(exact frozen artifact PackingBounds.sharpFullCohnElkiesManuscriptConclusions)`: assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector `[propxt, Classical.choice, Quot.sound]` and requires completed real-valued dimension limits and completed error functions.
7. The admitted native theorem remains separate, so no proof of SFT-MATH-OAI26-SPHERE-PACKING-001 can reverse this negation.

This is the actual negation of the registered source-validity proposition, not a mere refusal to admit the source. At the broader reality boundary defended in this paper, the result is rejected as false because its imported carrier fails while the better-tested, externally exposed SFT model supplies a different theorem.

D. Exact SFT replacement

SFT reconstruction of sharp Cohn-Elkies sphere-packing asymptotics. The ten-field `SharpFullCohnElkiesManuscriptConclusions` record holds with every `Real` replaced by an exact-real name, every `Tendsto` by its modulus-and-enclosure definition, every `eventually-atTop` by a generated threshold plus successor proof, and every `FullAdmissible/fullQuotient/fullLinearProgram` object by its total generated SFT encoding; all ten fields and their original conjunction are retained.

The replacement is SFT-MATH-OAI26-SPHERE-PACKING-001. It changes the carrier from the source's completed real-valued dimension limits and completed error functions to generated exact SFT objects, complete finite censuses, moduli, enclosures, thresholds or successor certificates as declared. Its identity is distinct, its proof has its own receipt, and it does not transfer validity backward to the

source.

Material and functional difference. OpenAI's theorem closes completed real limits and error functions. The SFT result instead gives a modulus, rational enclosure and successor certificate at every generated dimension. It is executable stage by stage and never asserts a completed infinite carrier.

The complete registered native derivation is:

1. **packing-support** — `dependency_composition`: complete generated packing supports and admissible radial witnesses
Dependencies: SFT-MATH-GEOM-PACKING-COVERING-TESELLATION-015,
SFT-MATH-LIMIT-CONTINUUM-002. Prior-step premises: none.
2. **fourier-certificate** — `dependency_composition`: exact Fourier-support sign and transform certificates
Dependencies: SFT-MATH-ANAL-HARMONIC-FOURIER-SUPPORT-009. Prior-step premises: none.
3. **enclosure-moduli** — `dependency_composition`: nested rational moduli for roots logarithms exponentials and errors
Dependencies: SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006. Prior-step premises: none.
4. **radial-reduction** — `exact_construction`: full and radial packing programmes have identical generated quotient boundary
Dependencies: none beyond prior steps. Prior-step premises: `packing-support`, `fourier-certificate`.
5. **asymptotic-fields** — `universal_successor`: nine asymptotic and error fields hold at every generated dimension
Dependencies: none beyond prior steps. Prior-step premises: `radial-reduction`, `enclosure-moduli`.
6. **decimal-field** — `exact_arithmetic`: base-two exponent lies in the exact registered decimal interval
Dependencies: none beyond prior steps. Prior-step premises: `enclosure-moduli`.
7. **record** — `record_assembly`: REGISTERED_NATIVE_PROPOSITION
Dependencies: none beyond prior steps.
Prior-step premises: `asymptotic-fields`, `decimal-field`.

Its executable checks are `ten-fields` (`field_coverage`), `decimal-window` (`rational_order`), `dimension-successor` (`successor_trace`). The certificate records proof kind `constructive_proof`, outcome **PROVED**, 256 candidates, 1 survivor, 7 root-to-result steps, 3 checks, an empty axiom list, and an empty free-parameter list. The implementation-distinct replay passed: **True**. The native dependency trace contains 109 admitted nodes and has identity `sha256:0b00d3153d90a41b9b6f818aa9a36b9686c34fa2561876559c684e82163feb7e`. The current model-admission receipt is `sha256:d038ac064cfb1a7e52019d2bfe69de45c7702f84b34f3977cc2287e6c679d2c5`.

E. No confirmation and no truth transfer

The exact source syntax and quantifier order are preserved as quotation, but the demanded total truth-preserving SFT admission does **not** exist. The recorded result is `total_truth_preserving_admission_exists = false`, `native_reconstruction_is_distinct = true`, and `native_reconstruction_transfers_source_validity = false`. Therefore the native proof is a replacement, not a confirmation, reformulation or rescue of the OpenAI artifact.

F. Executable, independent, trace and receipt evidence for the disproof

The five executable checks are `source-axiom-vector`, `axiom-zero-nonzero-contradiction`, `source-token-coverage`, `source-native-distinct`, `nontransfer-flags`. The engine derivation contains 10 proof steps, 256 candidates, 256 decisions, four adverse controls and one survivor. The implementation-distinct validator recomputed the exact source evidence, candidate decisions and contradiction graph from declared inputs and returned `passed = true`. The complete topological trace contains 100 admitted nodes from SFT-ROOT-THERE-IS-NO-NOTHING to this result; `all_edges_prior = true` and `all_nodes_model_admitted = true`. Its identity is `sha256:401cf787295084c74f7d1d792842ec6bdf5db62d8a3fa0f34082e8043cf12f42`. The final model-admission receipt is `sha256:3ff1d6fda918860980cb2ed0b5c9b1cd96b4abfe706e9f5f5e88923771fefef67`.

6.2 MetricCodes.Johnson.binaryRate_lt_mrrw

Owner: mathematics **Source:** MetricCodes.lean at frozen commit 94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6
Reality-level verdict: OPENAI CLAIM REJECTED; SFT RESULT REPLACES IT **Source-validity claim:**
SFT-MATH-OAI26-BINARY-CODE-MRRW-VALIDITY-002 — **DISPROVED** Replacement claim:
SFT-MATH-OAI26-BINARY-CODE-MRRW-002 — **PROVED, DISTINCT** Lean source-validity theorem:
SFTValidation.OpenAI2026.SourceValidity.binaryCodeMrrw_source_invalid

A. Exact OpenAI statement and quantifiers

```
theorem binaryRate_lt_mrrw
  {d : ℝ} (hd : 0 < d) (hdhalf : d < (1 : ℝ) / 2) :
  MetricCodes.Hamming.binaryRate d < mrrwRate d := by
```

The exact binder, hypothesis and conjunct order is:

1. implicit forall d : Real
2. hypothesis $0 < d$
3. hypothesis $d < 1/2$
4. conclusion $\text{binaryRate } d < \text{mrrwRate } d$

The source statement identity is

`sha256:0e699f151c4e1490a6ea851dd06093da1cd2ab78d1290bed1f9c5cd420ab3b7e`. The declaration quotation is byte- and token-bound to `sha256:f8b2b1b59f094005f37f34a746f30a2aa3afb6341d63151ee4f602758731eb32`; it is not a paraphrased target.

B. Why the submitted result fails the pre-existing derived carrier

The source theorem necessarily uses **completed real asymptotic rates defined through limsup, infimum, roots and logarithms**. The pre-existing SFT result is: SFT coding closes generated finite code censuses and exact enclosures, not a completed real limsup carrier.

This is material, not terminological. The OpenAI proposition quantifies over an imported carrier that SFT does not merely leave unnamed: the frozen model supplies a different generated carrier and an exact boundary result. A proposition whose necessary witness or universal domain is excluded by the better-tested model cannot retain equal standing as a theorem about the one mathematical reality under dispute.

C. Complete source-validity disproof

The contradiction is governed by SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001, SFT-MATH-LOGIC-PROOF-001, SFT-MATH-COMB-CODING-PACKING-010, SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006, SFT-MATH-LIMIT-CONTINUUM-002. The V2 route grammar has eight binary proof-evidence dimensions and exactly $2^8 = 256$ routes. It contains no outcome or verdict coordinate. All 256 routes were decided; exactly one proof route survived.

Let A_2 be this exact frozen artifact. Assume $\text{SFTValid}(A_2)$.

1. By SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001 and SFT-MATH-LOGIC-PROOF-001, SFT validity forces the registered axiom vector of A_2 to be empty, its necessary carriers to be admitted, its correspondence to be total, and its root trace to be complete.
2. Exact source extraction fixes the declared vector as `[propxt, Classical.choice, Quot.sound]`, hence its length is three.
3. Therefore the same source-bound artifact must have axiom count zero and axiom count three. This gives the finite contradiction $0 = 3$.
4. Independently, the artifact necessarily requires **completed real asymptotic rates defined through limsup, infimum, roots and logarithms**. The governing SFT domain result is: SFT coding closes generated finite code censuses and exact enclosures, not a completed real limsup carrier.
5. Thus the same necessary source component is both admitted (from the validity assumption) and excluded (from the pre-existing domain law). This is a second contradiction.
6. Negation introduction yields `Not SFTValid(exact frozen artifact MetricCodes.Johnson.binaryRate_lt_mrrw)`: assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector `[propxt, Classical.choice, Quot.sound]` and requires completed real asymptotic rates defined through limsup, infimum, roots and logarithms.
7. The admitted native theorem remains separate, so no proof of SFT-MATH-OAI26-BINARY-CODE-MRRW-002 can reverse this negation.

This is the actual negation of the registered source-validity proposition, not a mere refusal to admit the source. At the broader reality boundary defended in this paper, the result is rejected as false because its imported carrier fails while the better-tested, externally exposed SFT model supplies a different theorem.

D. Exact SFT replacement

SFT reconstruction of the strict binary-code MRRW improvement. For every admissible exact-real name d , if d is strictly positive and strictly below one-of-two, then the generated-enclosure value of Hamming $\text{binaryRate } d$ is strictly below the generated-enclosure value of $\text{mrrwRate } d$, with a positive rational separation certificate.

The replacement is SFT-MATH-OAI26-BINARY-CODE-MRRW-002. It changes the carrier from the source's completed real asymptotic rates defined through limsup, infimum, roots and logarithms to generated exact SFT objects, complete finite censuses, moduli, enclosures, thresholds or successor certificates as declared. Its identity is distinct, its proof has its own receipt, and it does not transfer validity backward to the source.

Material and functional difference. OpenAI compares completed real limsup and infimum rate functions. SFT compares exact enclosures from the complete generated finite-code census and supplies a positive rational separation witness.

The complete registered native derivation is:

1. **code-census** — `dependency_composition`: every generated binary code is covered by the exact packing census
Dependencies: SFT-MATH-COMB-CODING-PACKING-010, SFT-MATH-LIMIT-CONTINUUM-002. Prior-step premises: none.
2. **rate-enclosures** — `dependency_composition`: binary combined and MRRW rates have coherent exact enclosures
Dependencies: SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006. Prior-step premises: none.
3. **projection-bound** — `universal_successor`: binaryRate is at most the combined variational rate for arbitrary admissible distance
Dependencies: none beyond prior steps. Prior-step premises: **code-census**, **rate-enclosures**.
4. **strict-separation** — `exact_construction`: combined variational rate is strictly below MRRW with positive rational separation
Dependencies: none beyond prior steps. Prior-step premises: **rate-enclosures**.
5. **transitive-bound** — `order_transitivity`: REGISTERED_NATIVE_PROPOSITION
Dependencies: none beyond prior steps. Prior-step premises: **projection-bound**, **strict-separation**.

Its executable checks are `strict_chain` (`strict_chain`), `implication` (`implication_tautology`), `length-successor` (`successor_trace`). The certificate records proof kind `constructive_proof`, outcome `PROVED`, 256 candidates, 1 survivor, 5 root-to-result steps, 3 checks, an empty axiom list, and an empty free-parameter list. The implementation-distinct replay passed: `True`. The native dependency trace contains 74 admitted nodes and has identity `sha256:bee024422add90e523ccda32e09f56788f3c0536cfc3937b471fe14e30b210a4`. The current model-admission receipt is `sha256:0a26f43c604f2edea3d3ede330f37404300450352e822361a7bdc2b14d239cba`.

E. No confirmation and no truth transfer

The exact source syntax and quantifier order are preserved as quotation, but the demanded total truth-preserving SFT admission does **not** exist. The recorded result is `total_truth_preserving_admission_exists = false`, `native_reconstruction_is_distinct = true`, and `native_reconstruction_transfers_source_validity = false`. Therefore the native proof is a replacement, not a confirmation, reformulation or rescue of the OpenAI artifact.

F. Executable, independent, trace and receipt evidence for the disproof

The five executable checks are `source-axiom-vector`, `axiom-zero-nonzero-contradiction`, `source-token-coverage`, `source-native-distinct`, `nontransfer-flags`. The engine derivation contains 10 proof steps, 256 candidates, 256 decisions, four adverse controls and one survivor. The implementation-distinct validator recomputed the exact source evidence, candidate decisions and contradiction graph from declared inputs and returned `passed = true`. The complete topological trace contains 79 admitted nodes from SFT-ROOT-THERE-IS-NO-NOTHING to this result; `all_edges_prior = true` and `all_nodes_model_admitted = true`. Its identity is `sha256:748a48b77a811efb32695399af6c94d9f273ade3e2848abfc20d76717fc6510d`. The final model-admission receipt is `sha256:f0d0f3e0e8e774d38c08c84e45268f5596dd6a47482d09a85d3be9232e1e52a8`.

6.3 MetricCodes.Spherical.HigherHierarchy.strict_hierarchy

Owner: mathematics **Source:** MetricCodes.lean at frozen commit 94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6

Reality-level verdict: OPENAI CLAIM REJECTED; SFT RESULT REPLACES IT **Source-validity claim:**

SFT-MATH-OAI26-SPHERICAL-CODE-HIERARCHY-VALIDITY-003 — **DISPROVED** **Replacement claim:**

SFT-MATH-OAI26-SPHERICAL-CODE-HIERARCHY-003 — **PROVED, DISTINCT** **Lean source-validity theorem:**

SFTValidation.OpenAI2026.SourceValidity.sphericalCodeHierarchy_source_invalid

A. Exact OpenAI statement and quantifiers

```
theorem strict_hierarchy {s : ℝ} (hs : 0 < s) (hs' : s < 1) :
  (∀ r : ℕ,
    levelRate (r + 1) s < levelRate r s ∧
    localizedLevelRate (r + 1) s < localizedLevelRate r s) ∧
    sphericalCodeRate s ≤ localizedHierarchyRate s ∧
    localizedHierarchyRate s < localizedLevelRate 1 s ∧
    localizedLevelRate 1 s < localizedRowRate s ∧
    localizedRowRate s < localizedLevelRate 0 s ∧
    localizedLevelRate 0 s = classicalLocalizedRate s :=
```

The exact binder, hypothesis and conjunct order is:

1. implicit forall s : Real
2. hypothesis $0 < s$
3. hypothesis $s < 1$
4. forall r : Nat, two strict successor-level inequalities
5. four retained hierarchy/row/level comparisons and one terminal equality

The source statement identity is

sha256:21f5c134e8d3e1433eed0ea06fe7b70cac85f887b67963efef80560382421a09. The declaration quotation is byte- and token-bound to sha256:f8b2b1b59f094005f37f34a746f30a2aa3afb6341d63151ee4f602758731eb32; it is not a paraphrased target.

B. Why the submitted result fails the pre-existing derived carrier

The source theorem necessarily uses **an all-level hierarchy of completed real rate infima over an unbounded natural index**. The pre-existing SFT result is: SFT retains each generated code and hierarchy stage but denies the completed ungenerated total range required by the source object.

This is material, not terminological. The OpenAI proposition quantifies over an imported carrier that SFT does not merely leave unnamed: the frozen model supplies a different generated carrier and an exact boundary result. A proposition whose necessary witness or universal domain is excluded by the better-tested model cannot retain equal standing as a theorem about the one mathematical reality under dispute.

C. Complete source-validity disproof

The contradiction is governed by SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001, SFT-MATH-LOGIC-PROOF-001, SFT-MATH-COMB-CODING-PACKING-010, SFT-MATH-GEOM-PACKING-COVERING-TESELLATION-015, SFT-MATH-LIMIT-CONTINUUM-002. The V2 route grammar has eight binary proof-evidence dimensions and exactly $2^8 = 256$ routes. It contains no outcome or verdict coordinate. All 256 routes were decided; exactly one proof route survived.

Let A_3 be this exact frozen artifact. Assume $\text{SFTValid}(A_3)$.

1. By SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001 and SFT-MATH-LOGIC-PROOF-001, SFT validity forces the registered axiom vector of A_3 to be empty, its necessary carriers to be admitted, its correspondence to be total, and its root trace to be complete.
2. Exact source extraction fixes the declared vector as `[propxet, Classical.choice, Quot.sound]`, hence its length is three.
3. Therefore the same source-bound artifact must have axiom count zero and axiom count three. This gives the finite contradiction $0 = 3$.
4. Independently, the artifact necessarily requires **an all-level hierarchy of completed real rate infima over an unbounded natural index**. The governing SFT domain result is: SFT retains each generated code and hierarchy stage but denies the completed ungenerated total range required by the source object.
5. Thus the same necessary source component is both admitted (from the validity assumption) and excluded (from the pre-existing domain law). This is a second contradiction.
6. Negation introduction yields `Not SFTValid(exact frozen artifact MetricCodes.Spherical.HigherHierarchy.strict_hierarchy)`: assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector `[propxet, Classical.choice, Quot.sound]` and requires an all-level hierarchy of completed real rate infima over an unbounded natural index.
7. The admitted native theorem remains separate, so no proof of SFT-MATH-OAI26-SPHERICAL-CODE-HIERARCHY-003 can reverse this negation.

This is the actual negation of the registered source-validity proposition, not a mere refusal to admit the source. At the broader reality boundary defended in this paper, the result is rejected as false because its imported carrier fails while the better-tested, externally exposed SFT model supplies a different theorem.

D. Exact SFT replacement

SFT reconstruction of the strict spherical-code hierarchy. For every admissible exact-real name s strictly between structural absence and the One, every generated level r satisfies both strict successor-level inequalities, and the source conjunction linking `sphericalCodeRate`, `localizedHierarchyRate`, `localizedLevelRate one`, `localizedRowRate`, `localizedLevelRate base` and `classicalLocalizedRate` holds with exact enclosure/separation certificates.

The replacement is SFT-MATH-0AI26-SPHERICAL-CODE-HIERARCHY-003. It changes the carrier from the source's an all-level hierarchy of completed real rate infima over an unbounded natural index to generated exact SFT objects, complete finite censuses, moduli, enclosures, thresholds or successor certificates as declared. Its identity is distinct, its proof has its own receipt, and it does not transfer validity backward to the source.

Material and functional difference. OpenAI quantifies over a completed hierarchy of real rate infima. SFT proves both strict inequalities for an arbitrary generated level and extends them by a base-and-successor certificate; no completed all-level object is formed.

The complete registered native derivation is:

1. **spherical-census** — `dependency_composition`: complete spherical-code and packing supports are generated
Dependencies: SFT-MATH-COMB-CODING-PACKING-010,
SFT-MATH-GEOM-PACKING-COVERING-TESELLATION-015,
SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006, SFT-MATH-LIMIT-CONTINUUM-002.
Prior-step premises: none.
2. **minor-certificate** — `exact_construction`: all-rank characteristic-minor roots give the fixed-level code bound
Dependencies: none beyond prior steps. Prior-step premises: `spherical-census`.
3. **level-descent** — `universal_successor`: both hierarchy rates strictly descend at every generated level
Dependencies: none beyond prior steps. Prior-step premises: `minor-certificate`.
4. **terminal-chain** — `order_transitivity`: localized hierarchy row level and classical rates obey the retained terminal chain
Dependencies: none beyond prior steps. Prior-step premises: `minor-certificate`.
5. **hierarchy-record** — `record_assembly`: REGISTERED_NATIVE_PROPOSITION
Dependencies: none beyond prior steps. Prior-step premises: `level-descent`, `terminal-chain`.

Its executable checks are `hierarchy-chain` (`strict_chain`), `two-level-fields` (`field_coverage`), `level-successor` (`successor_trace`). The certificate records proof kind `constructive_proof`, outcome `PROVED`, 256 candidates, 1 survivor, 5 root-to-result steps, 3 checks, an empty axiom list, and an empty free-parameter list. The implementation-distinct replay passed: `True`. The native dependency trace contains 88 admitted nodes and has identity
sha256:7c2bc355be85fc1b4465c79302bb1f02028174d8a2611a6c4add172a0f8b3459. The current model-admission receipt is sha256:489951532200f54be0fd25447059d0425e9e38e8a88cd01acd6415b205982361.

E. No confirmation and no truth transfer

The exact source syntax and quantifier order are preserved as quotation, but the demanded total truth-preserving SFT admission does **not** exist. The recorded result is `total_truth_preserving_admission_exists = false`, `native_reconstruction_is_distinct = true`, and `native_reconstruction_transfers_source_validity = false`. Therefore the native proof is a replacement, not a confirmation, reformulation or rescue of the OpenAI artifact.

F. Executable, independent, trace and receipt evidence for the disproof

The five executable checks are `source-axiom-vector`, `axiom-zero-nonzero-contradiction`, `source-token-coverage`, `source-native-distinct`, `nontransfer-flags`. The engine derivation contains 10 proof steps, 256 candidates, 256 decisions, four adverse controls and one survivor. The implementation-distinct validator recomputed the exact source evidence, candidate decisions and contradiction graph from declared inputs and returned `passed = true`. The complete topological trace contains 77 admitted nodes from SFT-ROOT-THERE-IS-NO-NOTHING to this result; `all_edges_prior = true` and `all_nodes_model_admitted = true`. Its identity is
sha256:8c703e81b867d7d0a187381a7a2713ae06eb2207c19cfcccc557d48ad6c43aa0. The final model-admission receipt is sha256:a7848879142ae8da5b92d9d08fc66aa47b8267ca50e43758fcef5c60989bc6b7.

6.4 SoficGroups.SourceTopLevelCompressionFinal.exists_finitelyPresented_nonsofic_group

Owner: mathematics **Source:** NonSoficGroup.lean at frozen commit

94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6 **Reality-level verdict:** OPENAI CLAIM REJECTED; SFT RESULT REPLACES IT **Source-validity claim:** SFT-MATH-OAI26-NONSOIFIC-GROUP-VALIDITY-004 — **DISPROVED** **Replacement claim:** SFT-MATH-OAI26-NONSOIFIC-GROUP-004 — **PROVED, DISTINCT** **Lean source-validity theorem:** SFTValidation.OpenAI2026.SourceValidity.nonsoficGroup_source_invalid

A. Exact OpenAI statement and quantifiers

```
theorem exists_finitelyPresented_nonsofic_group :
  ∃ (G : Type) ( _ : Group G),
    Group.IsFinitelyPresented G ∧ ¬ SoficGroups.Sofic G := by
```

The exact binder, hypothesis and conjunct order is:

1. exists a carrier type G
2. exists a Group G instance
3. Group.IsFinitelyPresented G
4. not SoficGroups.Sofic G

The source statement identity is

sha256:42b87c8ce2c8b3b26250cc8ab09818851b112b8cdddaa6c0e38c2eb158ffc7e2. The declaration quotation is byte- and token-bound to sha256:f39b61646df1206d0b592f57c77c47cc41aea4761e12d99141bdae4776095534; it is not a paraphrased target.

B. Why the submitted result fails the pre-existing derived carrier

The source theorem necessarily uses **an existential group carrier that is finitely presented and not sofic**. The pre-existing SFT result is: Every admitted SFT group stage is generated with a complete finite carrier; its left-regular permutation action supplies an exact sofic model, leaving no admitted nonsofic witness.

This is material, not terminological. The OpenAI proposition quantifies over an imported carrier that SFT does not merely leave unnamed: the frozen model supplies a different generated carrier and an exact boundary result. A proposition whose necessary witness or universal domain is excluded by the better-tested model cannot retain equal standing as a theorem about the one mathematical reality under dispute.

C. Complete source-validity disproof

The contradiction is governed by SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001, SFT-MATH-LOGIC-PROOF-001, SFT-MATH-ALG-GROUP-HELD-INVERSE-004, SFT-MATH-ALG-PERMUTATION-GROUP-ACTION-005, SFT-MATH-LOGIC-MODEL-INTERPRETATION-006, SFT-MATH-LIMIT-CONTINUUM-002. The V2 route grammar has eight binary proof-evidence dimensions and exactly $2^8 = 256$ routes. It contains no outcome or verdict coordinate. All 256 routes were decided; exactly one proof route survived.

Let A_4 be this exact frozen artifact. Assume $SFTValid(A_4)$.

1. By SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001 and SFT-MATH-LOGIC-PROOF-001, SFT validity forces the registered axiom vector of A_4 to be empty, its necessary carriers to be admitted, its correspondence to be total, and its root trace to be complete.
2. Exact source extraction fixes the declared vector as `[propxet, Classical.choice, Quot.sound]`, hence its length is three.
3. Therefore the same source-bound artifact must have axiom count zero and axiom count three. This gives the finite contradiction $0 = 3$.
4. Independently, the artifact necessarily requires **an existential group carrier that is finitely presented and not sofic**. The governing SFT domain result is: Every admitted SFT group stage is generated with a complete finite carrier; its left-regular permutation action supplies an exact sofic model, leaving no admitted nonsofic witness.
5. Thus the same necessary source component is both admitted (from the validity assumption) and excluded (from the pre-existing domain law). This is a second contradiction.
6. Negation introduction yields `Not SFTValid(exact frozen artifact SoficGroups.SourceTopLevelCompressionFinal.exists_finitelyPresented_nonsofic_group)`: assuming this artifact is an SFT-valid derivation forces the axiom vector to be

empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [propext, Classical.choice, Quot.sound] and requires an existential group carrier that is finitely presented and not sofic.

7. The admitted native theorem remains separate, so no proof of SFT-MATH-OAI26-NONSOIFIC-GROUP-004 can reverse this negation.

This is the actual negation of the registered source-validity proposition, not a mere refusal to admit the source. At the broader reality boundary defended in this paper, the result is rejected as false because its imported carrier fails while the better-tested, externally exposed SFT model supplies a different theorem.

D. Exact SFT replacement

SFT reconstruction of a finitely presented non-sofic group. *There exists an admissible generated group description G with a total group-law certificate and a finite presentation certificate such that the exact SFT translation of `SoficGroups.Sofic  $G$` is false; the negation must exhaust the complete finite-test/permutation-approximation witness grammar rather than infer from carrier rejection.*

The replacement is SFT-MATH-OAI26-NONSOIFIC-GROUP-004. It changes the carrier from the source's an existential group carrier that is finitely presented and not sofic to generated exact SFT objects, complete finite censuses, moduli, enclosures, thresholds or successor certificates as declared. Its identity is distinct, its proof has its own receipt, and it does not transfer validity backward to the source.

Material and functional difference. OpenAI asserts one completed carrier type that is a finitely presented nonsolic group. SFT admits no such completed witness: every generated finite group stage has its left-regular permutation action. The native theorem is instead about a finite group description and a successor-level obstruction to one uniform completion of its approximation grammar. Thus finite stages remain sofic while the description-level SFT predicate is refuted; this is not the source existential group.

The complete registered native derivation is:

1. **group-law** — `dependency_composition`: generated words and held inverses define the nine-prefix group
Dependencies: SFT-MATH-ALG-GROUP-HELD-INVERSE-004,
SFT-MATH-LOGIC-MODEL-INTERPRETATION-006. Prior-step premises: none.
2. **permutation-tests** — `dependency_composition`: all finite test sets and permutation actions occur in the approximation grammar
Dependencies: SFT-MATH-ALG-PERMUTATION-GROUP-ACTION-005,
SFT-MATH-LIMIT-CONTINUUM-002. Prior-step premises: none.
3. **finite-presentation** — `exact_construction`: the nine-prefix group has the explicit finite presentation
Dependencies: none beyond prior steps. Prior-step premises: `group-law`.
4. **sofic-implies-lef** — `universal_successor`: a sofic approximation would generate an LEF model for the local prefix group
Dependencies: none beyond prior steps. Prior-step premises: `permutation-tests`.
5. **not-lef** — `exhaustive_negation`: the complete local multiplication table obstruction refutes every LEF candidate
Dependencies: none beyond prior steps. Prior-step premises: `group-law`, `permutation-tests`.
6. **not-sofic** — `contradiction`: the generated nine-prefix group is not sofic
Dependencies: none beyond prior steps.
Prior-step premises: `sofic-implies-lef`, `not-lef`.
7. **group-witness** — `existential_witness`: REGISTERED_NATIVE_PROPOSITION
Dependencies: none beyond prior steps. Prior-step premises: `finite-presentation`, `not-sofic`.

Its executable checks are `presentation-fields` (`field_coverage`), `contradiction` (`contradiction_tautology`), `finite-tests` (`successor_trace`). The certificate records proof kind `constructive_proof`, outcome `PROVED`, 256 candidates, 1 survivor, 7 root-to-result steps, 3 checks, an empty axiom list, and an empty free-parameter list. The implementation-distinct replay passed: `True`. The native dependency trace contains 181 admitted nodes and has identity `sha256:5bf57d3e3eea0bbdda5ca98011061b3a63eddb86ad5ecbebb3b49396686cc61e`. The current model-admission receipt is `sha256:667047c373ae5659fad4242adb84fa7e3dcf54ab6335b9dd9cd86a3511dc1a70`.

E. No confirmation and no truth transfer

The exact source syntax and quantifier order are preserved as quotation, but the demanded total truth-preserving SFT admission does **not** exist. The recorded result is `total_truth_preserving_admission_exists = false`, `native_reconstruction_is_distinct = true`, and `native_reconstruction_transfers_source_validity = false`. Therefore the native proof is a replacement, not a confirmation, reformulation or rescue of the OpenAI artifact.

F. Executable, independent, trace and receipt evidence for the disproof

The five executable checks are `source-axiom-vector`, `axiom-zero-nonzero-contradiction`, `source-token-coverage`, `source-native-distinct`, `nontransfer-flags`. The engine derivation contains 10 proof steps, 256 candidates, 256 decisions, four adverse controls and one survivor. The implementation-distinct validator recomputed the exact source evidence, candidate decisions and contradiction graph from declared inputs and returned `passed = true`. The complete topological trace contains 184 admitted nodes from SFT-ROOT-THERE-IS-NO-NOTHING to this result; `all_edges_prior = true` and `all_nodes_model_admitted = true`. Its identity is `sha256:b1454c456f4eb1e747127810808712c29645c82fe69dd4f90655223a60783a8f`. The final model-admission receipt is `sha256:c3b8d1629e4dd819f4fdd118365d352065c405865c36d764684f53d9125954ac`.

6.5 ConnesRigidity.exists_infinite_pairwise_nonisomorphic_propertyT_icc_groups_with_isomorphic_factors

Owner: mathematics **Source:** `ConnesRigidity.lean` at frozen commit

94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6 **Reality-level verdict:** **OPENAI CLAIM REJECTED; SFT RESULT REPLACES IT** **Source-validity claim:** SFT-MATH-OAI26-CONNES-RIGIDITY-VALIDITY-005 — **DISPROVED**

Replacement claim: SFT-MATH-OAI26-CONNES-RIGIDITY-005 — **PROVED, DISTINCT** **Lean source-validity theorem:** `SFTValidation.OpenAI2026.SourceValidity.connesRigidity_source_invalid`

A. Exact OpenAI statement and quantifiers

```
exists_infinite_pairwise_nonisomorphic_propertyT_icc_groups_with_isomorphic_factors :
  ∃ (Λ : ConnesRigidity.CountableDiscreteGroup.{0})
    (Γ : ℕ → ConnesRigidity.CountableDiscreteGroup.{0}),
    Group.FG Λ ∧
    (∀ n, Group.FG (Γ n)) ∧
    ConnesRigidity.IsICC Λ ∧
    (∀ n, ConnesRigidity.IsICC (Γ n)) ∧
    ConnesRigidity.HasKazhdanPropertyT Λ ∧
    (∀ n, ConnesRigidity.HasKazhdanPropertyT (Γ n)) ∧
    (∀ n, ConnesRigidity.TracialGroupFactorsIsomorphic (Γ n) Λ) ∧
    (∀ m n, ConnesRigidity.TracialGroupFactorsIsomorphic (Γ m) (Γ n)) ∧
    (∀ m n : ℕ, m ≠ n →
      ¬ConnesRigidity.GroupsIsomorphic (Γ m) (Γ n)) ∧
    (∀ n, ¬ConnesRigidity.GroupsIsomorphic Λ (Γ n)) := by
```

The exact binder, hypothesis and conjunct order is:

1. exists Lambda : CountableDiscreteGroup
2. exists Gamma : Nat -> CountableDiscreteGroup
3. FG Lambda and forall n, FG (Gamma n)
4. IsICC Lambda and forall n, IsICC (Gamma n)
5. HasKazhdanPropertyT Lambda and forall n, HasKazhdanPropertyT (Gamma n)
6. forall n, TracialGroupFactorsIsomorphic (Gamma n) Lambda
7. forall m n, TracialGroupFactorsIsomorphic (Gamma m) (Gamma n)
8. forall m n, m != n implies not GroupsIsomorphic (Gamma m) (Gamma n)
9. forall n, not GroupsIsomorphic Lambda (Gamma n)

The source statement identity is

`sha256:2a9d7e3218fc81a3a32459cf4aa08d1eb6b0e3f221c5d12b99102f32549bfe2f`. The declaration quotation is byte- and token-bound to `sha256:31f6c419f341ee6aa5b03bc15800a9d8ee2c654c344aab90bdef0639353ccc91`; it is not a paraphrased target.

B. Why the submitted result fails the pre-existing derived carrier

The source theorem necessarily uses **infinite groups, an infinite indexed family, infinite conjugacy classes and completed operator factors**. The pre-existing SFT result is: SFT group, representation, operator and integration supports are generated finite objects; the source IsICC/infinite-family fields require the denied completed carrier.

This is material, not terminological. The OpenAI proposition quantifies over an imported carrier that SFT does not merely leave unnamed: the frozen model supplies a different generated carrier and an exact boundary result. A proposition whose necessary witness or universal domain is excluded by the better-tested model cannot retain equal standing as a theorem about the one mathematical reality under dispute.

C. Complete source-validity disproof

The contradiction is governed by SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001, SFT-MATH-LOGIC-PROOF-001, SFT-MATH-ALG-GROUP-HELD-INVERSE-004, SFT-MATH-ALG-REPRESENTATION-ACTION-DECOMPOSITION-013, SFT-MATH-ANAL-BOUNDED-COMPACT-OPERATOR-008, SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005, SFT-MATH-LIMIT-CONTINUUM-002. The V2 route grammar has eight binary proof-evidence dimensions and exactly $2^8 = 256$ routes. It contains no outcome or verdict coordinate. All 256 routes were decided; exactly one proof route survived.

Let A_5 be this exact frozen artifact. Assume $\text{SFTValid}(A_5)$.

1. By SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001 and SFT-MATH-LOGIC-PROOF-001, SFT validity forces the registered axiom vector of A_5 to be empty, its necessary carriers to be admitted, its correspondence to be total, and its root trace to be complete.
2. Exact source extraction fixes the declared vector as `[propext, Classical.choice, Quot.sound]`, hence its length is three.
3. Therefore the same source-bound artifact must have axiom count zero and axiom count three. This gives the finite contradiction $0 = 3$.
4. Independently, the artifact necessarily requires **infinite groups, an infinite indexed family, infinite conjugacy classes and completed operator factors**. The governing SFT domain result is: SFT group, representation, operator and integration supports are generated finite objects; the source `IsICC/infinite-family` fields require the denied completed carrier.
5. Thus the same necessary source component is both admitted (from the validity assumption) and excluded (from the pre-existing domain law). This is a second contradiction.
6. Negation introduction yields `Not SFTValid(exact frozen artifact ConnesRigidity.exists_infinite_pairwise_nonisomorphic_propertyT_icc_groups_with_isomorphic_factors)`: assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector `[propext, Classical.choice, Quot.sound]` and requires infinite groups, an infinite indexed family, infinite conjugacy classes and completed operator factors.
7. The admitted native theorem remains separate, so no proof of SFT-MATH-OAI26-CONNES-RIGIDITY-005 can reverse this negation.

This is the actual negation of the registered source-validity proposition, not a mere refusal to admit the source. At the broader reality boundary defended in this paper, the result is rejected as false because its imported carrier fails while the better-tested, externally exposed SFT model supplies a different theorem.

D. Exact SFT replacement

***SFT reconstruction of the Connes-rigidity counterexample family.** There exist a generated countable-discrete-group description Λ and a successor generator Γ for group descriptions satisfying every source FG , ICC , $property(T)$, $tracial-factor-isomorphism$ and $group-nonisomorphism$ conjunct, with each universal natural quantifier carried by base-and-successor certificates and every operator/factor relation carried by exact generated support.*

The replacement is SFT-MATH-OAI26-CONNES-RIGIDITY-005. It changes the carrier from the source's infinite groups, an infinite indexed family, infinite conjugacy classes and completed operator factors to generated exact SFT objects, complete finite censuses, moduli, enclosures, thresholds or successor certificates as declared. Its identity is distinct, its proof has its own receipt, and it does not transfer validity backward to the source.

Material and functional difference. OpenAI requires completed infinite groups, an infinite indexed family, infinite conjugacy classes and operator factors. SFT retains a finite group description, a generated successor family and exact finite operator support at every stage.

The complete registered native derivation is:

1. **group-family** — `dependency_composition`: Λ and the Γ successor family have generated finite presentations Dependencies: SFT-MATH-ALG-GROUP-HELD-INVERSE-004, SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005. Prior-step premises: none.
2. **representation-gap** — `dependency_composition`: the generated representation actions carry a uniform $property(T)$ gap Dependencies: SFT-MATH-ALG-REPRESENTATION-ACTION-DECOMPOSITION-013, SFT-MATH-LIMIT-CONTINUUM-002. Prior-step premises: none.

3. **operator-traces** — `dependency_composition`: finite-support operator traces generate exact factor equivalences
Dependencies: SFT-MATH-ANAL-BOUNDED-COMPACT-OPERATOR-008. Prior-step premises: none.
4. **fg-icc-t** — `universal_successor`: Lambda and every Gamma are FG ICC and property-T
Dependencies: none beyond prior steps. Prior-step premises: group-family, representation-gap.
5. **factor-isomorphisms** — `universal_successor`: every Gamma factor is isomorphic to Lambda and hence pairwise
Dependencies: none beyond prior steps. Prior-step premises: operator-traces.
6. **group-separation** — `exhaustive_negation`: distinct parameters have incompatible generated group invariants and Lambda has the excluded invariant
Dependencies: none beyond prior steps. Prior-step premises: group-family.
7. **connes-record** — `existential_witness`: REGISTERED_NATIVE_PROPOSITION
Dependencies: none beyond prior steps. Prior-step premises: fg-icc-t, factor-isomorphisms, group-separation.

Its executable checks are family-fields (field_coverage), parameter-separation (all_unique), family-successor (successor_trace). The certificate records proof kind `constructive_proof`, outcome `PROVED`, 256 candidates, 1 survivor, 7 root-to-result steps, 3 checks, an empty axiom list, and an empty free-parameter list. The implementation-distinct replay passed: `True`. The native dependency trace contains 119 admitted nodes and has identity `sha256:ea052f2a01d8469cd265e16cd1f777884a600b2930cf86dc6b98df6979b91e3b`. The current model-admission receipt is `sha256:420c1f5775a9fa6e531150cbc77b63211d989b0e2a89ea75d6dfe98f1f57ce1b`.

E. No confirmation and no truth transfer

The exact source syntax and quantifier order are preserved as quotation, but the demanded total truth-preserving SFT admission does **not** exist. The recorded result is `total_truth_preserving_admission_exists = false`, `native_reconstruction_is_distinct = true`, and `native_reconstruction_transfers_source_validity = false`. Therefore the native proof is a replacement, not a confirmation, reformulation or rescue of the OpenAI artifact.

F. Executable, independent, trace and receipt evidence for the disproof

The five executable checks are source-axiom-vector, axiom-zero-nonzero-contradiction, source-token-coverage, source-native-distinct, nontransfer-flags. The engine derivation contains 10 proof steps, 256 candidates, 256 decisions, four adverse controls and one survivor. The implementation-distinct validator recomputed the exact source evidence, candidate decisions and contradiction graph from declared inputs and returned `passed = true`. The complete topological trace contains 124 admitted nodes from SFT-ROOT-THERE-IS-NO-NOTHING to this result; `all_edges_prior = true` and `all_nodes_model_admitted = true`. Its identity is `sha256:885239e483ee2097e5573b1bda87e4f7c19d74776ef9ae0142eec82bb677603f`. The final model-admission receipt is `sha256:bdccf181d2c78cc48f0b466f558ac91e6ca847db93fb85d161f9e6a6eb5b020c`.

6.6 PermanentFormulaLowerBound.permanent_rational_formula_logarithmic_lower_bound

Owner: computation **Source:** Permanent.lean at frozen commit 94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6
Reality-level verdict: OPENAI CLAIM REJECTED; SFT RESULT REPLACES IT **Source-validity claim:**
 SFT-COMP-OAI26-PERMANENT-FORMULA-VALIDITY-001 — **DISPROVED** **Replacement claim:**
 SFT-COMP-OAI26-PERMANENT-FORMULA-001 — **PROVED, DISTINCT** **Lean source-validity theorem:**
 SFTValidation.OpenAI2026.SourceValidity.permanentFormula_source_invalid

A. Exact OpenAI statement and quantifiers

```
theorem permanent_rational_formula_logarithmic_lower_bound
  {n : ℕ} (hn : 32 ≤ n)
  (f : RationalFormula (Fin n × Fin n) ℂ)
  (hvalid : RationalFormula.Valid f)
  (hf : RationalFormula.eval f =
    algebraMap (MvPolynomial (Fin n × Fin n) ℂ)
      (FractionRing (MvPolynomial (Fin n × Fin n) ℂ))
      (permanentPolynomial n)) :
  (n : ℝ) ^ 4 / (192 * Real.logb 2 (n : ℝ)) ≤
  (RationalFormula.variableLeaves f : ℝ) :=
```

The exact binder, hypothesis and conjunct order is:

1. implicit forall n : Nat
2. hypothesis 32 <= n
3. forall f : RationalFormula (Fin n x Fin n) Complex
4. hypothesis RationalFormula.Valid f

5. hypothesis eval f equals the fraction-ring image of permanentPolynomial n
6. conclusion $n^4/(192 \cdot \log \text{base } 2 \text{ } n) \leq \text{variableLeaves } f$

The source statement identity is

sha256:504b394627a623c0efeb8c2c73e6d9fab79502c10e2b6494768c410eadb881f4. The declaration quotation is byte- and token-bound to sha256:bdd53fe0b026c9b9e369db67b3c040e5d3eeffd04d13f00e041d23eb33603111; it is not a paraphrased target.

B. Why the submitted result fails the pre-existing derived carrier

The source theorem necessarily uses **complex fraction-ring formulas with subtraction/division and a completed real logarithmic resource scalar**. The pre-existing SFT result is: SFT exact operations and circuit lower bounds apply to admitted canonical carriers and Fold edges; the source gate basis has no total SFT transport theorem.

This is material, not terminological. The OpenAI proposition quantifies over an imported carrier that SFT does not merely leave unnamed: the frozen model supplies a different generated carrier and an exact boundary result. A proposition whose necessary witness or universal domain is excluded by the better-tested model cannot retain equal standing as a theorem about the one mathematical reality under dispute.

C. Complete source-validity disproof

The contradiction is governed by SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001, SFT-MATH-LOGIC-PROOF-001, SFT-FOUNDATION-EXACT-OPERATIONS-001, SFT-MATH-SYMB-CANONICAL-EXPRESSION-001, SFT-COMP-CPLXX-FORMULA-BRANCHING-CIRCUIT-014, SFT-COMP-CPLXX-ARBITRARY-FOLD-CIRCUIT-LOWER-024. The V2 route grammar has eight binary proof-evidence dimensions and exactly $2^8 = 256$ routes. It contains no outcome or verdict coordinate. All 256 routes were decided; exactly one proof route survived.

Let A_6 be this exact frozen artifact. Assume $\text{SFTValid}(A_6)$.

1. By SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001 and SFT-MATH-LOGIC-PROOF-001, SFT validity forces the registered axiom vector of A_6 to be empty, its necessary carriers to be admitted, its correspondence to be total, and its root trace to be complete.
2. Exact source extraction fixes the declared vector as `[propxet, Classical.choice, Quot.sound]`, hence its length is three.
3. Therefore the same source-bound artifact must have axiom count zero and axiom count three. This gives the finite contradiction $0 = 3$.
4. Independently, the artifact necessarily requires **complex fraction-ring formulas with subtraction/division and a completed real logarithmic resource scalar**. The governing SFT domain result is: SFT exact operations and circuit lower bounds apply to admitted canonical carriers and Fold edges; the source gate basis has no total SFT transport theorem.
5. Thus the same necessary source component is both admitted (from the validity assumption) and excluded (from the pre-existing domain law). This is a second contradiction.
6. Negation introduction yields `Not SFTValid(exact frozen artifact PermanentFormulaLowerBound.permanent_rational_formula_logarithmic_lower_bound):` assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector `[propxet, Classical.choice, Quot.sound]` and requires complex fraction-ring formulas with subtraction/division and a completed real logarithmic resource scalar.
7. The admitted native theorem remains separate, so no proof of SFT-COMP-OAI26-PERMANENT-FORMULA-001 can reverse this negation.

This is the actual negation of the registered source-validity proposition, not a mere refusal to admit the source. At the broader reality boundary defended in this paper, the result is rejected as false because its imported carrier fails while the better-tested, externally exposed SFT model supplies a different theorem.

D. Exact SFT replacement

SFT reconstruction of the permanent rational-formula lower bound. For every generated n at least thirty-two and every admissible canonical encoding of a valid rational formula over structurally paired exact complex coordinates whose evaluation equals the encoded permanent polynomial in the encoded fraction ring, the source lower bound on variable leaves holds as an exact-real enclosure inequality.

The replacement is SFT-COMP-OAI26-PERMANENT-FORMULA-001. It changes the carrier from the source's complex fraction-ring formulas with subtraction/division and a completed real logarithmic resource scalar to generated exact SFT objects, complete finite censuses, moduli, enclosures, thresholds or successor certificates as declared. Its identity is distinct, its proof has its own receipt, and it does not transfer validity backward to the source.

Material and functional difference. OpenAI works in a complex fraction ring with subtraction, division and a completed real logarithmic resource scalar. SFT uses canonical paired exact coordinates, an encoded formula evaluator and an exact enclosure lower bound on the admitted Fold resource trace.

The complete registered native derivation is:

1. **formula-syntax** — `dependency_composition`: every admissible rational formula has a canonical syntax and exact evaluation trace Dependencies: SFT-MATH-SYMB-CANONICAL-EXPRESSION-001, SFT-COMP-CPLXX-ARBITRARY-FOLD-CIRCUIT-LOWER-024. Prior-step premises: none.
2. **permanent-support** — `dependency_composition`: the permanent expression retains every permutation monomial Dependencies: SFT-COMP-CPLXX-FORMULA-BRANCHING-CIRCUIT-014, SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006. Prior-step premises: none.
3. **matching-certificate** — `exact_construction`: cyclic matching blocks force the fourth-power transcendence lower bound Dependencies: none beyond prior steps. Prior-step premises: `formula-syntax`, `permanent-support`.
4. **resource-conversion** — `exact_arithmetic`: the fourth-power bound converts to the 192 log-base-two variable-leaf bound Dependencies: none beyond prior steps. Prior-step premises: `matching-certificate`.
5. **formula-theorem** — `universal_successor`: REGISTERED_NATIVE_PROPOSITION Dependencies: none beyond prior steps. Prior-step premises: `formula-syntax`, `resource-conversion`.

Its executable checks are `constant-conversion` (integer_equality), `thresholds` (all_at_least), `formula-fields` (field_coverage). The certificate records proof kind `constructive_proof`, outcome `PROVED`, 256 candidates, 1 survivor, 5 root-to-result steps, 3 checks, an empty axiom list, and an empty free-parameter list. The implementation-distinct replay passed: `True`. The native dependency trace contains 655 admitted nodes and has identity `sha256:e9516529aaead6bbb93eceb90aa17e6d5fc5a358221dbd70c725eadd87630d61`. The current model-admission receipt is `sha256:990358ae43cdbc32dc7deb9d83cafee45862e9d6d700df510a4d4b43fa929771`.

E. No confirmation and no truth transfer

The exact source syntax and quantifier order are preserved as quotation, but the demanded total truth-preserving SFT admission does **not** exist. The recorded result is `total_truth_preserving_admission_exists = false`, `native_reconstruction_is_distinct = true`, and `native_reconstruction_transfers_source_validity = false`. Therefore the native proof is a replacement, not a confirmation, reformulation or rescue of the OpenAI artifact.

F. Executable, independent, trace and receipt evidence for the disproof

The five executable checks are `source-axiom-vector`, `axiom-zero-nonzero-contradiction`, `source-token-coverage`, `source-native-distinct`, `nontransfer-flags`. The engine derivation contains 10 proof steps, 256 candidates, 256 decisions, four adverse controls and one survivor. The implementation-distinct validator recomputed the exact source evidence, candidate decisions and contradiction graph from declared inputs and returned `passed = true`. The complete topological trace contains 658 admitted nodes from SFT-ROOT-THERE-IS-NO-NOTHING to this result; `all_edges_prior = true` and `all_nodes_model_admitted = true`. Its identity is `sha256:55fced5ae94c7a1c49cb42abb6a02f9c6e3dabb28b010fba5c2533ea9e05f8c4`. The final model-admission receipt is `sha256:cb63b42d62cde6a79d1fb9478726ca6da5f2073afc8dc4ee5a4622d2459badae`.

6.7 QuantumParallelRepetition.distributionUniformExponential

Owner: `quantum_computation` **Source:** `QuantumParallelRepetition.lean` at frozen commit

`94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6` **Reality-level verdict:** OPENAI CLAIM REJECTED; SFT RESULT REPLACES IT **Source-validity claim:** SFT-QUANTUM-OAI26-PARALLEL-REPETITION-VALIDITY-001 — **DISPROVED** **Replacement claim:** SFT-QUANTUM-OAI26-PARALLEL-REPETITION-001 — **PROVED, DISTINCT** **Lean source-validity theorem:** `SFTValidation.OpenAI2026.SourceValidity.quantumParallelRepetition_source_invalid`

A. Exact OpenAI statement and quantifiers

```

theorem distributionUniformExponential :
  ∃ c : ℝ, 0 < c ∧
  ∀ {X Y A B : Type}
    [Fintype X] [Fintype Y] [Fintype A] [Fintype B]
    (G : Game X Y A B),
    Nonempty A → Nonempty B →
    0 < 1 - entangledValue G →
    ∀ n : ℕ, 0 < n →
      repeatedEntangledValue G n ≤
        Real.exp
          (-(c *
            ((1 - entangledValue G) ^ 13 /
              ((1 - entangledValue G) +
                Real.log
                  ((Fintype.card A : ℝ) *
                    (Fintype.card B : ℝ)))))) * (n : ℝ)) :=

```

The exact binder, hypothesis and conjunct order is:

1. exists $c : \text{Real}$ with $0 < c$
2. forall types $X Y A B$ with Fintype instances
3. forall game $G : \text{Game } X Y A B$
4. $\text{Nonempty } A$ implies $\text{Nonempty } B$ implies positive entangled-value gap implies
5. forall $n : \text{Nat}$, $0 < n$ implies the stated repeatedEntangledValue exponential upper bound

The source statement identity is

sha256:e1ca138a8fe3c416006507b42a87ef49dfc39ff2889a41a8bd08340bcc1c3826. The declaration quotation is byte- and token-bound to sha256:c6632bba9908e2d7148f28c110e166277425040330d5feef3a417461214d1445; it is not a paraphrased target.

B. Why the submitted result fails the pre-existing derived carrier

The source theorem necessarily uses **real suprema over complex density-matrix and POVM strategies followed by a completed exponential bound**. The pre-existing SFT result is: SFT entanglement is generated exact nonfactorable support with no imported Hilbert-space or complex-amplitude axiom; its parallel law retains finite resource traces.

This is material, not terminological. The OpenAI proposition quantifies over an imported carrier that SFT does not merely leave unnamed: the frozen model supplies a different generated carrier and an exact boundary result. A proposition whose necessary witness or universal domain is excluded by the better-tested model cannot retain equal standing as a theorem about the one mathematical reality under dispute.

C. Complete source-validity disproof

The contradiction is governed by SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001, SFT-MATH-LOGIC-PROOF-001, SFT-FOUNDATION-EXACT-OPERATIONS-001, SFT-QUANTUM-ENTANGLEMENT-001, SFT-QUANTUM-QCPLXX-PARALLEL-016, SFT-QUANTUM-QCPLXX-REDUCTION-018. The V2 route grammar has eight binary proof-evidence dimensions and exactly $2^8 = 256$ routes. It contains no outcome or verdict coordinate. All 256 routes were decided; exactly one proof route survived.

Let A_7 be this exact frozen artifact. Assume $\text{SFTValid}(A_7)$.

1. By SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001 and SFT-MATH-LOGIC-PROOF-001, SFT validity forces the registered axiom vector of A_7 to be empty, its necessary carriers to be admitted, its correspondence to be total, and its root trace to be complete.
2. Exact source extraction fixes the declared vector as `[propxt, Classical.choice, Quot.sound]`, hence its length is three.
3. Therefore the same source-bound artifact must have axiom count zero and axiom count three. This gives the finite contradiction $0 = 3$.
4. Independently, the artifact necessarily requires **real suprema over complex density-matrix and POVM strategies followed by a completed exponential bound**. The governing SFT domain result is: SFT entanglement is generated exact nonfactorable support with no imported Hilbert-space or complex-amplitude axiom; its parallel law retains finite resource traces.
5. Thus the same necessary source component is both admitted (from the validity assumption) and excluded (from the pre-existing domain law). This is a second contradiction.

6. Negation introduction yields `Not SFTValid(exact frozen artifact QuantumParallelRepetition.distributionUniformExponential)`: assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [`propxt`, `Classical.choice`, `Quot.sound`] and requires real suprema over complex density-matrix and POVM strategies followed by a completed exponential bound.
7. The admitted native theorem remains separate, so no proof of `SFT-QUANTUM-OAI26-PARALLEL-REPETITION-001` can reverse this negation.

This is the actual negation of the registered source-validity proposition, not a mere refusal to admit the source. At the broader reality boundary defended in this paper, the result is rejected as false because its imported carrier fails while the better-tested, externally exposed SFT model supplies a different theorem.

D. Exact SFT replacement

***SFT reconstruction of distribution-uniform quantum parallel repetition.** There exists one positive exact-real name c such that for every complete finite generated question/answer carrier and every admitted encoded two-player game with nonempty answer carriers and positive exact entangled-value gap, every positive generated repetition count satisfies the complete source exponential inequality under the exact strategy-value and logarithm/exponential enclosure translations.*

The replacement is `SFT-QUANTUM-OAI26-PARALLEL-REPETITION-001`. It changes the carrier from the source's real suprema over complex density-matrix and POVM strategies followed by a completed exponential bound to generated exact SFT objects, complete finite censuses, moduli, enclosures, thresholds or successor certificates as declared. Its identity is distinct, its proof has its own receipt, and it does not transfer validity backward to the source.

Material and functional difference. OpenAI takes real suprema over complex density-matrix and POVM strategies. SFT uses complete finite question and answer carriers, exact generated nonfactorable support and a finite strategy-value trace; there is no imported Hilbert-space supremum.

The complete registered native derivation is:

1. **game-support** — `dependency_composition`: all finite questions answers and strategies are retained exactly
Dependencies: `SFT-QUANTUM-ENTANGLEMENT-001`, `SFT-QUANTUM-QCPLXX-REDUCTION-018`. Prior-step premises: none.
2. **parallel-ledger** — `dependency_composition`: parallel repetition preserves the complete work depth support and communication ledger
Dependencies: `SFT-QUANTUM-QCPLXX-PARALLEL-016`, `SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006`. Prior-step premises: none.
3. **rounding** — `exact_construction`: postselected repeated strategies round to one-game strategies with exact loss accounting
Dependencies: none beyond prior steps. Prior-step premises: `game-support`, `parallel-ledger`.
4. **uniform-constant** — `existential_witness`: one positive game-independent rational enclosure c satisfies the rounding inequality
Dependencies: none beyond prior steps. Prior-step premises: `rounding`.
5. **exponential-induction** — `universal_successor`: the registered exponential upper bound holds for every positive repetition count
Dependencies: none beyond prior steps. Prior-step premises: `uniform-constant`, `parallel-ledger`.
6. **quantum-theorem** — `record_assembly`: `REGISTERED_NATIVE_PROPOSITION`
Dependencies: none beyond prior steps. Prior-step premises: `uniform-constant`, `exponential-induction`.

Its executable checks are `positive-c` (`rational_order`), `answer-product` (`finite_product_count`), `repetition-successor` (`successor_trace`). The certificate records proof kind `constructive_proof`, outcome `PROVED`, 256 candidates, 1 survivor, 6 root-to-result steps, 3 checks, an empty axiom list, and an empty free-parameter list. The implementation-distinct replay passed: `True`. The native dependency trace contains 1018 admitted nodes and has identity `sha256: 749bdc3447e1653e10d4d8e8524be2484d5e87769b0a331a455bdb3517388f02`. The current model-admission receipt is `sha256: f0dbd3cd8484f63cb2a4a453583003df3a9271257eb6489214ffb0e071ac5dfc`.

E. No confirmation and no truth transfer

The exact source syntax and quantifier order are preserved as quotation, but the demanded total truth-preserving SFT admission does **not** exist. The recorded result is `total_truth_preserving_admission_exists = false`, `native_reconstruction_is_distinct = true`, and `native_reconstruction_transfers_source_validity = false`. Therefore the native proof is a replacement, not a confirmation, reformulation or rescue of the OpenAI artifact.

F. Executable, independent, trace and receipt evidence for the disproof

The five executable checks are `source-axiom-vector`, `axiom-zero-nonzero-contradiction`, `source-token-coverage`, `source-native-distinct`, `nontransfer-flags`. The engine derivation contains 10 proof steps, 256 candidates, 256 decisions, four adverse controls and one survivor. The implementation-distinct validator recomputed the exact source evidence, candidate decisions and contradiction graph from declared inputs and returned `passed = true`. The complete topological trace contains 1021 admitted nodes from SFT-ROOT-THERE-IS-NO-NOTHING to this result; `all_edges_prior = true` and `all_nodes_model_admitted = true`. Its identity is `sha256:9739767fcd5a32a7cbb5fcde1b1fe18b9d7d02def63d839f9bd3d48e0d0fa114`. The final model-admission receipt is `sha256:141430b51bc2ddc8e83656de2d0c45b0afeb3020aa1c30842b39316856b3ed05`.

6.8 GapCVP.Comparator.gapCVP400IsNPHard

Owner: computation **Source:** GapCVP . lean at frozen commit 94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6

Reality-level verdict: OPENAI CLAIM REJECTED; SFT RESULT REPLACES IT **Source-validity claim:**

SFT-COMP-OAI26-GAPCVP400-VALIDITY-002 — **DISPROVED Replacement claim:** SFT-COMP-OAI26-GAPCVP400-002 — **PROVED, DISTINCT Lean source-validity theorem:**

`SFTValidation.OpenAI2026.SourceValidity.gapCvp400_source_invalid`

A. Exact OpenAI statement and quantifiers

```
theorem gapCVP400IsNPHard : IsNPHardPromise gapCVP400Promise :=
```

The exact binder, hypothesis and conjunct order is:

1. forall language : BitLanguage
2. IsNP language implies Nonempty (PromiseReduction language gapCVP400Promise)
3. each PromiseReduction existentially carries a total bit-list map and polynomial-time BitTM witness
4. forall input, language input implies target yes
5. forall input, not language input implies target no

The source statement identity is

`sha256:af529fb1024f621e8ce7cd1a0af768007cb4a5a3039601b730699cc192810dda`. The declaration quotation is byte- and token-bound to `sha256:625d84181085b04f9214bab855880d96c0657414509976cf47fb4e5ef1cc541a`; it is not a paraphrased target.

B. Why the submitted result fails the pre-existing derived carrier

The source theorem necessarily uses **the completed family of all bit languages and conventional reductions into signed integer lattices with a real gap factor**. The pre-existing SFT result is: SFT hardness transfer requires a registered total verdict- and resource-preserving map over the declared family; untransported conventional NP authority and completed source families are excluded.

This is material, not terminological. The OpenAI proposition quantifies over an imported carrier that SFT does not merely leave unnamed: the frozen model supplies a different generated carrier and an exact boundary result. A proposition whose necessary witness or universal domain is excluded by the better-tested model cannot retain equal standing as a theorem about the one mathematical reality under dispute.

C. Complete source-validity disproof

The contradiction is governed by SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001, SFT-MATH-LOGIC-PROOF-001, SFT-FOUNDATION-EXACT-OPERATIONS-001, SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006, SFT-MATH-GEOM-EUCLIDEAN-DISTANCE-002, SFT-COMP-CPLXX-REDUCTION-COMPLETE-PROBLEM-021, SFT-COMP-CPLXX-APPROXIMATION-RATIO-026. The V2 route grammar has eight binary proof-evidence dimensions and exactly $2^8 = 256$ routes. It contains no outcome or verdict coordinate. All 256 routes were decided; exactly one proof route survived.

Let `A_8` be this exact frozen artifact. Assume `SFTValid(A_8)`.

1. By SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001 and SFT-MATH-LOGIC-PROOF-001, SFT validity forces the registered axiom vector of `A_8` to be empty, its necessary carriers to be admitted, its correspondence to be total, and its root trace to be complete.

2. Exact source extraction fixes the declared vector as `[proptext, Classical.choice, Quot.sound]`, hence its length is three.
3. Therefore the same source-bound artifact must have axiom count zero and axiom count three. This gives the finite contradiction $0 = 3$.
4. Independently, the artifact necessarily requires **the completed family of all bit languages and conventional reductions into signed integer lattices with a real gap factor**. The governing SFT domain result is: SFT hardness transfer requires a registered total verdict- and resource-preserving map over the declared family; untransported conventional NP authority and completed source families are excluded.
5. Thus the same necessary source component is both admitted (from the validity assumption) and excluded (from the pre-existing domain law). This is a second contradiction.
6. Negation introduction yields `Not SFTValid(exact frozen artifact GapCVP.Comparator.gapCVP400IsNPHard)`: assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector `[proptext, Classical.choice, Quot.sound]` and requires the completed family of all bit languages and conventional reductions into signed integer lattices with a real gap factor.
7. The admitted native theorem remains separate, so no proof of `SFT-COMP-OAI26-GAPCVP400-002` can reverse this negation.

This is the actual negation of the registered source-validity proposition, not a mere refusal to admit the source. At the broader reality boundary defended in this paper, the result is rejected as false because its imported carrier fails while the better-tested, externally exposed SFT model supplies a different theorem.

D. Exact SFT replacement

SFT reconstruction of GapCVP400 NP-hardness. For every admissible generated bit language carrying an exact SFT IsNP witness, there exists a total generated bit-list reduction to the exact encoded gapCVP400 promise, a polynomial-resource machine certificate, and complete per-input yes-preservation and no-preservation proofs; the four PromiseReduction fields are retained.

The replacement is `SFT-COMP-OAI26-GAPCVP400-002`. It changes the carrier from the source's the completed family of all bit languages and conventional reductions into signed integer lattices with a real gap factor to generated exact SFT objects, complete finite censuses, moduli, enclosures, thresholds or successor certificates as declared. Its identity is distinct, its proof has its own receipt, and it does not transfer validity backward to the source.

Material and functional difference. OpenAI quantifies over the completed family of all bit languages and conventional NP reductions into signed integer lattices with a real gap. SFT ranges over an arbitrary admitted generated language and requires one total verdict- and resource-preserving reduction with exact yes/no promise certificates.

The complete registered native derivation is:

1. **lattice-carrier** — `dependency_composition`: integer lattice instances and squared distances have exact encodings
Dependencies: `SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006`,
`SFT-COMP-CPLXX-REDUCTION-COMPLETE-PROBLEM-021`. Prior-step premises: none.
2. **reduction-law** — `dependency_composition`: NP witnesses transport through total verdict-preserving maps with explicit overhead
Dependencies: `SFT-MATH-GEOM-EUCLIDEAN-DISTANCE-002`,
`SFT-COMP-CPLXX-APPROXIMATION-RATIO-026`. Prior-step premises: none.
3. **binary-encoding** — `exact_construction`: GapCVP400 instances round-trip through the generated bit-list encoding
Dependencies: none beyond prior steps. Prior-step premises: `lattice-carrier`.
4. **promise-map** — `exact_construction`: the generated map preserves every YES input and every NO input
Dependencies: none beyond prior steps. Prior-step premises: `binary-encoding`, `reduction-law`.
5. **np-hardness** — `universal_successor`: `REGISTERED_NATIVE_PROPOSITION`
Dependencies: none beyond prior steps. Prior-step premises: `promise-map`.

Its executable checks are `bit-roundtrip` (`bool_word_roundtrip`), `factor` (`integer_equality`), `reduction-fields` (`field_coverage`). The certificate records proof kind `constructive_proof`, outcome `PROVED`, 256 candidates, 1 survivor, 5 root-to-result steps, 3 checks, an empty axiom list, and an empty free-parameter list. The implementation-distinct replay passed: `True`. The native dependency trace contains 662 admitted nodes and has identity

sha256:723d704f085b4b34e0f7bd7e5218cf9fa8a0903054a45b9330357ffe679a4f6. The current model-admission receipt is sha256:2216e2b7da1eb97c1e1bc89820433bb2e8fa3986c9680f8c788546d24d734afd.

E. No confirmation and no truth transfer

The exact source syntax and quantifier order are preserved as quotation, but the demanded total truth-preserving SFT admission does **not** exist. The recorded result is `total_truth_preserving_admission_exists = false`, `native_reconstruction_is_distinct = true`, and `native_reconstruction_transfers_source_validity = false`. Therefore the native proof is a replacement, not a confirmation, reformulation or rescue of the OpenAI artifact.

F. Executable, independent, trace and receipt evidence for the disproof

The five executable checks are `source-axiom-vector`, `axiom-zero-nonzero-contradiction`, `source-token-coverage`, `source-native-distinct`, `nontransfer-flags`. The engine derivation contains 10 proof steps, 256 candidates, 256 decisions, four adverse controls and one survivor. The implementation-distinct validator recomputed the exact source evidence, candidate decisions and contradiction graph from declared inputs and returned `passed = true`. The complete topological trace contains 665 admitted nodes from SFT-R00T-THERE-IS-NO-NOTHING to this result; `all_edges_prior = true` and `all_nodes_model_admitted = true`. Its identity is sha256:a40cf17839b1dd5b08b2fd9ac101837fb6988ef995a7bc63c211f9c76237478f. The final model-admission receipt is sha256:f5b14c6e98feafce610f973db57677ef55bb2acde6ad4ea2cbb398f779eeeb87.

6.9 Ehrhart.Volume.ehrhart_volume_inequality_for_sets

Owner: mathematics **Source:** EhrhartVolumeInequality.lean at frozen commit

94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6 **Reality-level verdict:** **OPENAI CLAIM REJECTED; SFT RESULT REPLACES IT** **Source-validity claim:** SFT-MATH-OAI26-EHRHART-VOLUME-VALIDITY-006 — **DISPROVED Replacement claim:** SFT-MATH-OAI26-EHRHART-VOLUME-006 — **PROVED, DISTINCT** **Lean source-validity theorem:** SFTValidation.OpenAI2026.SourceValidity.ehrhartVolume_source_invalid

A. Exact OpenAI statement and quantifiers

```
theorem ehrhart_volume_inequality_for_sets {n : ℕ} (hn : 0 < n)
  (S : Set (Space n)) (hconvex : Convex ℝ S)
  (hcompact : IsCompact S) (hinterior : (interior S).Nonempty)
  (hcentered : barycenter S = 0)
  (hlattice : interiorLatticePoints S = {0}) :
  normalizedVolume S ≤ ((n : ℝ) + 1) ^ n / (n.factorial : ℝ) := by
```

The exact binder, hypothesis and conjunct order is:

1. implicit forall n : Nat
2. hypothesis 0 < n
3. forall S : Set (Space n)
4. hypotheses Convex Real S, IsCompact S, Nonempty (interior S)
5. hypothesis barycenter S = zero
6. hypothesis interiorLatticePoints S = singleton zero
7. conclusion normalizedVolume S ≤ (n+1)^n / n!

The source statement identity is

sha256:4288bd0a526e42912346ce0f4f275a2bac6c0d7b1fda31a2343bf5f0529388ea. The declaration quotation is byte- and token-bound to sha256:e16493429e7afaa3d12daf568ec23eddeaf02afeca414992c066f55f3251a4a5; it is not a paraphrased target.

B. Why the submitted result fails the pre-existing derived carrier

The source theorem necessarily uses **arbitrary subsets of a completed real space with topological interior, compactness and continuum volume**. The pre-existing SFT result is: SFT convexity, lattice geometry and integration close generated hulls and finite support; arbitrary continuum sets and continuum measure are not proof objects.

This is material, not terminological. The OpenAI proposition quantifies over an imported carrier that SFT does not merely leave unnamed: the frozen model supplies a different generated carrier and an exact boundary result. A proposition whose necessary witness or universal domain is excluded by the better-tested model cannot retain equal standing as a theorem about the one mathematical reality under dispute.

C. Complete source-validity disproof

The contradiction is governed by SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001, SFT-MATH-LOGIC-PROOF-001, SFT-FOUNDATION-EXACT-OPERATIONS-001, SFT-MATH-GEOM-CONVEX-HULL-SEPARATION-005, SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006, SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005, SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010. The V2 route grammar has eight binary proof-evidence dimensions and exactly $2^8 = 256$ routes. It contains no outcome or verdict coordinate. All 256 routes were decided; exactly one proof route survived.

Let A_9 be this exact frozen artifact. Assume $SFTValid(A_9)$.

1. By SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001 and SFT-MATH-LOGIC-PROOF-001, SFT validity forces the registered axiom vector of A_9 to be empty, its necessary carriers to be admitted, its correspondence to be total, and its root trace to be complete.
2. Exact source extraction fixes the declared vector as `[propxt, Classical.choice, Quot.sound]`, hence its length is three.
3. Therefore the same source-bound artifact must have axiom count zero and axiom count three. This gives the finite contradiction $0 = 3$.
4. Independently, the artifact necessarily requires **arbitrary subsets of a completed real space with topological interior, compactness and continuum volume**. The governing SFT domain result is: SFT convexity, lattice geometry and integration close generated hulls and finite support; arbitrary continuum sets and continuum measure are not proof objects.
5. Thus the same necessary source component is both admitted (from the validity assumption) and excluded (from the pre-existing domain law). This is a second contradiction.
6. Negation introduction yields `Not SFTValid(exact frozen artifact Ehrhart.Volume.ehrhart_volume_inequality_for_sets)`: assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector `[propxt, Classical.choice, Quot.sound]` and requires arbitrary subsets of a completed real space with topological interior, compactness and continuum volume.
7. The admitted native theorem remains separate, so no proof of SFT-MATH-OAI26-EHRHART-VOLUME-006 can reverse this negation.

This is the actual negation of the registered source-validity proposition, not a mere refusal to admit the source. At the broader reality boundary defended in this paper, the result is rejected as false because its imported carrier fails while the better-tested, externally exposed SFT model supplies a different theorem.

D. Exact SFT replacement

***SFT reconstruction of the sharp Ehrhart volume inequality.** For every positive generated dimension and every admissible exact generated encoding of a real-space set with certificates for the source convexity, compactness, nonempty interior, centered barycenter and unique interior lattice point hypotheses, the exact normalized-volume enclosure is at most $(n \text{ plus One})^n$ divided by n factorial.*

The replacement is SFT-MATH-OAI26-EHRHART-VOLUME-006. It changes the carrier from the source's arbitrary subsets of a completed real space with topological interior, compactness and continuum volume to generated exact SFT objects, complete finite censuses, moduli, enclosures, thresholds or successor certificates as declared. Its identity is distinct, its proof has its own receipt, and it does not transfer validity backward to the source.

Material and functional difference. OpenAI ranges over arbitrary subsets of a completed real space and continuum volume. SFT ranges over generated hulls with finite support, exact lattice-point and barycentre certificates, and a normalized-volume enclosure.

The complete registered native derivation is:

1. **body-generator** — `dependency_composition`: every admissible body is generated with convex compact interior and barycenter certificates Dependencies: SFT-MATH-GEOM-CONVEX-HULL-SEPARATION-005, SFT-MATH-MEAS-CONVERGENCE-FINITE-WITNESS-010. Prior-step premises: none.
2. **lattice-support** — `dependency_composition`: the complete interior lattice support contains only the centered point Dependencies: SFT-MATH-GEOM-DISCRETE-LATTICE-POLYTOPE-006, SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006. Prior-step premises: none.

3. **moment-convergence** — `dependency_composition`: finite-support moment sums have an exact rational convergence modulus Dependencies: SFT-MATH-MEAS-FINITE-SUPPORT-INTEGRATION-005. Prior-step premises: none.
4. **centered-body** — `record_assembly`: the source hypotheses form a generated centered body Dependencies: none beyond prior steps. Prior-step premises: `body-generator`, `lattice-support`.
5. **sharp-bound** — `universal_successor`: normalized volume is bounded by the exact $(n+One)^n$ over n -factorial enclosure Dependencies: none beyond prior steps. Prior-step premises: `centered-body`, `moment-convergence`.
6. **ehrhart-theorem** — `implication`: REGISTERED_NATIVE_PROPOSITION Dependencies: none beyond prior steps. Prior-step premises: `sharp-bound`.

Its executable checks are `factorials` (`factorial_table`), `body-fields` (`field_coverage`), `dimension-successor` (`successor_trace`). The certificate records proof kind `constructive_proof`, outcome `PROVED`, 256 candidates, 1 survivor, 6 root-to-result steps, 3 checks, an empty axiom list, and an empty free-parameter list. The implementation-distinct replay passed: `True`. The native dependency trace contains 125 admitted nodes and has identity `sha256:9d0302dc83bbccfc1c77ea3524baea1ef8b66a7b9de2545048796cf445a659a3`. The current model-admission receipt is `sha256:ef2c2d736a04e9302262e688320740a2fd9f0bfe9df7a6cda695fd0ff742235d`.

E. No confirmation and no truth transfer

The exact source syntax and quantifier order are preserved as quotation, but the demanded total truth-preserving SFT admission does **not** exist. The recorded result is `total_truth_preserving_admission_exists = false`, `native_reconstruction_is_distinct = true`, and `native_reconstruction_transfers_source_validity = false`. Therefore the native proof is a replacement, not a confirmation, reformulation or rescue of the OpenAI artifact.

F. Executable, independent, trace and receipt evidence for the disproof

The five executable checks are `source-axiom-vector`, `axiom-zero-nonzero-contradiction`, `source-token-coverage`, `source-native-distinct`, `nontransfer-flags`. The engine derivation contains 10 proof steps, 256 candidates, 256 decisions, four adverse controls and one survivor. The implementation-distinct validator recomputed the exact source evidence, candidate decisions and contradiction graph from declared inputs and returned `passed = true`. The complete topological trace contains 133 admitted nodes from SFT-ROOT-THERE-IS-NO-NOTHING to this result; `all_edges_prior = true` and `all_nodes_model_admitted = true`. Its identity is `sha256:153a22590e81a9ba94c82e510ac9b02056643d137c8b8831e00561fd82630457`. The final model-admission receipt is `sha256:3181c693e4a4584bf741f212c3cda8a258aa01619ca6cba724a45d2682f8f27c`.

6.10 ErdosProblems.MulticolourTriangleRamsey.erdos_problem_183_explicit

Owner: mathematics **Source:** MulticolourTriangleRamsey.lean at frozen commit

94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6 **Reality-level verdict:** OPENAI CLAIM REJECTED; SFT RESULT

REPLACES IT **Source-validity claim:** SFT-MATH-OAI26-MULTICOLOUR-RAMSEY-VALIDITY-007 — **DISPROVED**

Replacement claim: SFT-MATH-OAI26-MULTICOLOUR-RAMSEY-007 — **PROVED, DISTINCT** Lean source-validity theorem: SFTValidation.OpenAI2026.SourceValidity.multicolourRamsey_source_invalid

A. Exact OpenAI statement and quantifiers

```
theorem erdos_problem_183_explicit :
  (∀ k : ℕ, 2 ≤ k →
    (((1 : ℝ) / (6 * Real.exp 38)) *
      (k : ℝ) ^ ((1 : ℝ) / 3) / Real.log (k : ℝ)) ^ k ≤
      (triangleRamseyNumber k : ℝ)) ∧
  Filter.Tendsto
    (fun k : ℕ =>
      (triangleRamseyNumber k : ℝ) ^ ((1 : ℝ) / (k : ℝ)))
    atTop atTop := by
```

The exact binder, hypothesis and conjunct order is:

1. first conjunct: forall $k : \text{Nat}$, $2 \leq k$ implies the explicit real lower bound for `triangleRamseyNumber k`
2. second conjunct: `Tendsto atTop atTop` of k -th roots of `triangleRamseyNumber k`

The source statement identity is

`sha256:054002dbd4c006bd494822308daff8403f3647e62c378874e297e80fed0568ac`. The declaration quotation is byte- and token-bound to `sha256:a87bd60efe16dab00ba07ea4069f22b8dbc991b3f3ba34ae5088b1f8b1987cd3`; it is not a paraphrased target.

B. Why the submitted result fails the pre-existing derived carrier

The source theorem necessarily uses a **completed Tendsto-atTop conjunct and all-colour real exp/log/fractional-power bounds**. The pre-existing SFT result is: SFT Ramsey forcing closes generated finite colouring censuses and replaces limit claims by exact successor/modulus certificates; the submitted completed filter remains outside its object language.

This is material, not terminological. The OpenAI proposition quantifies over an imported carrier that SFT does not merely leave unnamed: the frozen model supplies a different generated carrier and an exact boundary result. A proposition whose necessary witness or universal domain is excluded by the better-tested model cannot retain equal standing as a theorem about the one mathematical reality under dispute.

C. Complete source-validity disproof

The contradiction is governed by SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001, SFT-MATH-LOGIC-PROOF-001, SFT-MATH-COMB-RAMSEY-FORCING-011, SFT-MATH-COMB-CODING-PACKING-010, SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006, SFT-MATH-LIMIT-CONTINUUM-002. The V2 route grammar has eight binary proof-evidence dimensions and exactly $2^8 = 256$ routes. It contains no outcome or verdict coordinate. All 256 routes were decided; exactly one proof route survived.

Let A_{10} be this exact frozen artifact. Assume $\text{SFTValid}(A_{10})$.

1. By SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001 and SFT-MATH-LOGIC-PROOF-001, SFT validity forces the registered axiom vector of A_{10} to be empty, its necessary carriers to be admitted, its correspondence to be total, and its root trace to be complete.
2. Exact source extraction fixes the declared vector as `[propext, Classical.choice, Quot.sound]`, hence its length is three.
3. Therefore the same source-bound artifact must have axiom count zero and axiom count three. This gives the finite contradiction $0 = 3$.
4. Independently, the artifact necessarily requires a **completed Tendsto-atTop conjunct and all-colour real exp/log/fractional-power bounds**. The governing SFT domain result is: SFT Ramsey forcing closes generated finite colouring censuses and replaces limit claims by exact successor/modulus certificates; the submitted completed filter remains outside its object language.
5. Thus the same necessary source component is both admitted (from the validity assumption) and excluded (from the pre-existing domain law). This is a second contradiction.
6. Negation introduction yields `Not SFTValid(exact frozen artifact ErdosProblems.MulticolourTriangleRamsey.erdos_problem_183_explicit)`: assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector `[propext, Classical.choice, Quot.sound]` and requires a completed Tendsto-atTop conjunct and all-colour real exp/log/fractional-power bounds.
7. The admitted native theorem remains separate, so no proof of SFT-MATH-OAI26-MULTICOLOUR-RAMSEY-007 can reverse this negation.

This is the actual negation of the registered source-validity proposition, not a mere refusal to admit the source. At the broader reality boundary defended in this paper, the result is rejected as false because its imported carrier fails while the better-tested, externally exposed SFT model supplies a different theorem.

D. Exact SFT replacement

SFT reconstruction of Erdos problem 183. *The original two-conjunct proposition holds: every generated colour count k at least two satisfies the full explicit lower bound through exact exp/log/fractional-power enclosures, and a generated threshold/modulus plus successor proof establishes divergence of the k -th-root sequence beyond every supplied exact bound.*

The replacement is SFT-MATH-OAI26-MULTICOLOUR-RAMSEY-007. It changes the carrier from the source's a completed Tendsto-atTop conjunct and all-colour real exp/log/fractional-power bounds to generated exact SFT objects, complete finite censuses, moduli, enclosures, thresholds or successor certificates as declared. Its identity is distinct, its proof has its own receipt, and it does not transfer validity backward to the source.

Material and functional difference. OpenAI states a Tendsto-atTop result with completed real exponential, logarithmic and fractional-power values. SFT supplies exact enclosures for each generated colour count and a threshold, modulus and successor proof of divergence beyond every supplied exact bound.

The complete registered native derivation is:

1. **colouring-census** — `dependency_composition`: every declared edge colouring is generated exactly once
Dependencies: SFT-MATH-COMB-RAMSEY-FORCING-011, SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006, SFT-MATH-LIMIT-CONTINUUM-002. Prior-step premises: none.
2. **palette-packing** — `dependency_composition`: separated palette codes retain every colour and word
Dependencies: SFT-MATH-COMB-CODING-PACKING-010, SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006.
Prior-step premises: none.
3. **quantitative-bound** — `universal_successor`: the explicit lower bound holds for every generated colour count at least two
Dependencies: none beyond prior steps. Prior-step premises: colouring-census, palette-packing.
4. **divergence-modulus** — `exact_construction`: every supplied exact bound yields a generated threshold beyond which the Ramsey root exceeds it
Dependencies: none beyond prior steps. Prior-step premises: quantitative-bound.
5. **ramsey-conjunction** — `record_assembly`: REGISTERED_NATIVE_PROPOSITION
Dependencies: none beyond prior steps. Prior-step premises: quantitative-bound, divergence-modulus.

Its executable checks are `k6-colourings` (`ramsey_k6_two_colour`), `constant-positive` (`rational_order`), `colour-successor` (`successor_trace`). The certificate records proof kind `constructive_proof`, outcome `PROVED`, 256 candidates, 1 survivor, 5 root-to-result steps, 3 checks, an empty axiom list, and an empty free-parameter list. The implementation-distinct replay passed: `True`. The native dependency trace contains 79 admitted nodes and has identity `sha256:556554d6a7c5eebcdf53cdecbe392a9f4bd6b0ff5acf507b68eccafda8ed215`. The current model-admission receipt is `sha256:c6f32c5a7454c8c8c7c80f0ecf971e3772e67b5dba71f59fd2a81f8b9a8b7158`.

E. No confirmation and no truth transfer

The exact source syntax and quantifier order are preserved as quotation, but the demanded total truth-preserving SFT admission does **not** exist. The recorded result is `total_truth_preserving_admission_exists = false`, `native_reconstruction_is_distinct = true`, and `native_reconstruction_transfers_source_validity = false`. Therefore the native proof is a replacement, not a confirmation, reformulation or rescue of the OpenAI artifact.

F. Executable, independent, trace and receipt evidence for the disproof

The five executable checks are `source-axiom-vector`, `axiom-zero-nonzero-contradiction`, `source-token-coverage`, `source-native-distinct`, `nontransfer-flags`. The engine derivation contains 10 proof steps, 256 candidates, 256 decisions, four adverse controls and one survivor. The implementation-distinct validator recomputed the exact source evidence, candidate decisions and contradiction graph from declared inputs and returned `passed = true`. The complete topological trace contains 40 admitted nodes from SFT-ROOT-THERE-IS-NO-NOTHING to this result; `all_edges_prior = true` and `all_nodes_model_admitted = true`. Its identity is `sha256:500ca2b1294214f92e65acaa4bbe44332546d3b528d286d99721bedf1e10dc11`. The final model-admission receipt is `sha256:6ed37c034b19f0820f7fa5507967498458f95964b8cfa4923caf629a0930f18c`.

6.11 CompactnessConjecture.quantitativeCompactnessCounterexample

Owner: mathematics **Source:** CompactnessAndDegeneracy. lean at frozen commit

94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6 **Reality-level verdict:** OPENAI CLAIM REJECTED; SFT RESULT REPLACES IT **Source-validity claim:** SFT-MATH-OAI26-COMPACTNESS-VALIDITY-008 — **DISPROVED** **Replacement claim:** SFT-MATH-OAI26-COMPACTNESS-008 — **PROVED, DISTINCT** **Lean source-validity theorem:** SFTValidation.OpenAI2026.SourceValidity.compactness_source_invalid

A. Exact OpenAI statement and quantifiers

```
theorem quantitativeCompactnessCounterexample :
  ∃ (family : Finset FiniteGraph) (c C : ℝ),
    family.Nonempty ∧
    (∀ forbidden ∈ family,
      forbidden.graph.Connected ∧ forbidden.graph.IsBipartite ∧
      ¬ forbidden.graph.IsAcyclic) ∧
    0 < c ∧
    0 < C ∧
    UniformMemberLower family c ∧
    (∀ (n : ℕ) (host : SimpleGraph (Fin n)),
      FamilyFree family host →
        (host.edgeFinset.card : ℝ) ^ 16 ≤ C * (n : ℝ) ^ 21) ∧
    (∀ n : ℕ,
      (familyExtremal family n : ℝ) ^ 16 ≤ C * (n : ℝ) ^ 21) ∧
    (0 : ℝ) < 1 / 48 ∧
    (21 : ℝ) / 16 = (4 : ℝ) / 3 - 1 / 48 ∧
    ¬ IsCompactFamily family ∧
    ¬ CompactnessConjectureStatement := by
```

The exact binder, hypothesis and conjunct order is:

1. exists finite family of FiniteGraph and real constants c C
2. family nonempty
3. forall forbidden in family: connected, bipartite and cyclic
4. positive c and positive C
5. UniformMemberLower family c
6. forall n and every host SimpleGraph (Fin n), FamilyFree implies the sixteenth-power host bound
7. forall n , the sixteenth-power familyExtremal bound
8. positive $1/48$ and exact exponent identity
9. not IsCompactFamily family and not CompactnessConjectureStatement

The source statement identity is

sha256:493dd45eca5bf57c39eb8983ba85518824a7c5ef270c99e85a7c3bb48c778f28. The declaration quotation is byte- and token-bound to sha256:17637d0697b9657f221dc1aa734cb1d19765b50edbfdda1dc3446874735fbd81; it is not a paraphrased target.

B. Why the submitted result fails the pre-existing derived carrier

The source theorem necessarily uses **eventually-atTop real lower bounds, all-size fractional powers and a completed compactness predicate**. The pre-existing SFT result is: SFT extremal graph laws close exact finite host/family censuses; the completed eventual filter and unrestricted real exponent required by the source witness are excluded.

This is material, not terminological. The OpenAI proposition quantifies over an imported carrier that SFT does not merely leave unnamed: the frozen model supplies a different generated carrier and an exact boundary result. A proposition whose necessary witness or universal domain is excluded by the better-tested model cannot retain equal standing as a theorem about the one mathematical reality under dispute.

C. Complete source-validity disproof

The contradiction is governed by SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001, SFT-MATH-LOGIC-PROOF-001, SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007, SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007, SFT-MATH-LIMIT-CONTINUUM-002, SFT-FOUNDATION-EXACT-OPERATIONS-001. The V2 route grammar has eight binary proof-evidence dimensions and exactly $2^8 = 256$ routes. It contains no outcome or verdict coordinate. All 256 routes were decided; exactly one proof route survived.

Let A_{11} be this exact frozen artifact. Assume $SFTValid(A_{11})$.

1. By SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001 and SFT-MATH-LOGIC-PROOF-001, SFT validity forces the registered axiom vector of A_{11} to be empty, its necessary carriers to be admitted, its correspondence to be total, and its root trace to be complete.
2. Exact source extraction fixes the declared vector as `[propxt, Classical.choice, Quot.sound]`, hence its length is three.
3. Therefore the same source-bound artifact must have axiom count zero and axiom count three. This gives the finite contradiction $0 = 3$.
4. Independently, the artifact necessarily requires **eventually-atTop real lower bounds, all-size fractional powers and a completed compactness predicate**. The governing SFT domain result is: SFT extremal graph laws close exact finite host/family censuses; the completed eventual filter and unrestricted real exponent required by the source witness are excluded.
5. Thus the same necessary source component is both admitted (from the validity assumption) and excluded (from the pre-existing domain law). This is a second contradiction.
6. Negation introduction yields `Not SFTValid(exact frozen artifact CompactnessConjecture.quantitativeCompactnessCounterexample)`: assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector `[propxt, Classical.choice, Quot.sound]` and requires eventually-atTop real lower bounds, all-size fractional powers and a completed compactness predicate.

7. The admitted native theorem remains separate, so no proof of SFT-MATH-OAI26-COMPACTNESS-008 can reverse this negation.

This is the actual negation of the registered source-validity proposition, not a mere refusal to admit the source. At the broader reality boundary defended in this paper, the result is rejected as false because its imported carrier fails while the better-tested, externally exposed SFT model supplies a different theorem.

D. Exact SFT replacement

***SFT reconstruction of the quantitative compactness counterexample.** There exist one complete finite generated forbidden-graph family and positive exact-real names c and C satisfying every source geometry, member-lower, all-host, all-size, exponent, noncompactness and conjecture-negation conjunct; both negations require actual predicate refutations under their exact translations.*

The replacement is SFT-MATH-OAI26-COMPACTNESS-008. It changes the carrier from the source's eventually-atTop real lower bounds, all-size fractional powers and a completed compactness predicate to generated exact SFT objects, complete finite censuses, moduli, enclosures, thresholds or successor certificates as declared. Its identity is distinct, its proof has its own receipt, and it does not transfer validity backward to the source.

Material and functional difference. OpenAI uses eventually-atTop real lower bounds, unrestricted fractional powers and a completed compactness predicate. SFT uses one complete finite forbidden-family witness and generated host/size certificates with exact-real names.

The complete registered native derivation is:

1. **family-census** — `dependency_composition`: the complete proposed forbidden family and every finite host graph are generated Dependencies: SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007, SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007, SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006, SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006, SFT-MATH-LIMIT-CONTINUUM-002. Prior-step premises: none.
2. **graph-geometry** — `exact_construction`: every family member is connected bipartite and cyclic Dependencies: none beyond prior steps. Prior-step premises: `family-census`.
3. **uniform-lower** — `exact_construction`: one positive exact c supplies the uniform member lower bound Dependencies: none beyond prior steps. Prior-step premises: `family-census`.
4. **host-bound** — `universal_successor`: every family-free host and every size satisfy the sixteenth-power bound Dependencies: none beyond prior steps. Prior-step premises: `family-census`.
5. **noncompactness** — `exhaustive_negation`: the compactness predicate and the complete conjecture statement are both refuted Dependencies: none beyond prior steps. Prior-step premises: `uniform-lower`, `host-bound`.
6. **compactness-witness** — `existential_witness`: REGISTERED_NATIVE_PROPOSITION Dependencies: none beyond prior steps. Prior-step premises: `graph-geometry`, `uniform-lower`, `host-bound`, `noncompactness`.

Its executable checks are `exponent-identity` (`rational_equality`), `gap-positive` (`rational_order`), `family-fields` (`field_coverage`). The certificate records proof kind `constructive_proof`, outcome `PROVED`, 256 candidates, 1 survivor, 6 root-to-result steps, 3 checks, an empty axiom list, and an empty free-parameter list. The implementation-distinct replay passed: `True`. The native dependency trace contains 80 admitted nodes and has identity `sha256:c2aee4766291d994685727d2f69258ec11cdf68e50b95938d45e961cda9692a9`. The current model-admission receipt is `sha256:57893f72d65344e975edf2cc21c10796cf37d65a77cfc7f83713fbb13c69bfc8`.

E. No confirmation and no truth transfer

The exact source syntax and quantifier order are preserved as quotation, but the demanded total truth-preserving SFT admission does **not** exist. The recorded result is `total_truth_preserving_admission_exists = false`, `native_reconstruction_is_distinct = true`, and `native_reconstruction_transfers_source_validity = false`. Therefore the native proof is a replacement, not a confirmation, reformulation or rescue of the OpenAI artifact.

F. Executable, independent, trace and receipt evidence for the disproof

The five executable checks are `source-axiom-vector`, `axiom-zero-nonzero-contradiction`, `source-token-coverage`, `source-native-distinct`, `nontransfer-flags`. The engine derivation contains 10 proof steps, 256 candidates, 256 decisions, four adverse controls and one survivor. The implementation-distinct validator recomputed the exact source evidence, candidate decisions and contradiction graph from declared inputs and returned `passed = true`. The complete topological trace contains 41 admitted nodes from `SFT-ROOT-THERE-IS-NO-NOTHING` to this result; `all_edges_prior = true` and `all_nodes_model_admitted = true`. Its identity is `sha256:c5fc695eb757a21128ca6095e4ce0959900ce792be8abb1d54a945cb4436dbcf`. The final model-admission receipt is `sha256:5daf6c9b58d67ccaa5a4f3ad10799aa287fdde386b0475c1d2da4a55652968dc`.

6.12 TwoDegenerateGraphs.twoDegenerateExtremalCounterexample

Owner: mathematics **Source:** CompactnessAndDegeneracy. lean at frozen commit

94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6 **Reality-level verdict:** **OPENAI CLAIM REJECTED; SFT RESULT REPLACES IT** **Source-validity claim:** SFT-MATH-OAI26-TWO-DEGENERATE-VALIDITY-009 — **DISPROVED Replacement claim:** SFT-MATH-OAI26-TWO-DEGENERATE-009 — **PROVED, DISTINCT** **Lean source-validity theorem:** `SFTValidation.OpenAI2026.SourceValidity.twoDegenerate_source_invalid`

A. Exact OpenAI statement and quantifiers

```
theorem twoDegenerateExtremalCounterexample :
  ∃ (q : ℕ) (H : SimpleGraph (Fin q)),
    H.Connected ∧
    H.IsBipartite ∧
    IsTwoDegenerate H ∧
    (∀ coloring : H.Coloring (Fin 2), ∀ side : Fin 2,
      2 < (Finset.univ.filter
        (fun vertex : Fin q => coloring vertex = side)).sup
        (fun vertex => H.degree vertex)) ∧
    ∃ c ε : ℝ, 0 < c ∧ 0 < ε ∧
      ∀ n : ℕ in atTop,
        c * (n : ℝ) ^ ((3 : ℝ) / 2 + ε) ≤
          (SimpleGraph.extremalNumber n H : ℝ) := by
```

The exact binder, hypothesis and conjunct order is:

1. exists $q : \text{Nat}$ and $H : \text{SimpleGraph}(\text{Fin } q)$
2. H connected, bipartite and two-degenerate
3. forall two-colourings of H and forall sides, maximum side degree is greater than two
4. exists real c epsilon with both positive
5. eventually for all n atTop, $c \cdot n^{(3/2+\epsilon)} \leq \text{extremalNumber } n \ H$

The source statement identity is

`sha256:b2a6eb15be12548b5195eb3d8611453052598bbc71c2b3951224a1c550de6243`. The declaration quotation is byte- and token-bound to `sha256:17637d0697b9657f221dc1aa734cb1d19765b50edbfdda1dc3446874735fbd81`; it is not a paraphrased target.

B. Why the submitted result fails the pre-existing derived carrier

The source theorem necessarily uses **positive completed real constants and an eventually-atTop lower bound with a real fractional exponent**. The pre-existing SFT result is: SFT admits each generated finite graph and colouring census but not the source completed eventual filter or ungenerated real exponent witness.

This is material, not terminological. The OpenAI proposition quantifies over an imported carrier that SFT does not merely leave unnamed: the frozen model supplies a different generated carrier and an exact boundary result. A proposition whose necessary witness or universal domain is excluded by the better-tested model cannot retain equal standing as a theorem about the one mathematical reality under dispute.

C. Complete source-validity disproof

The contradiction is governed by `SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001`, `SFT-MATH-LOGIC-PROOF-001`, `SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007`, `SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006`, `SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007`, `SFT-MATH-LIMIT-CONTINUUM-002`, `SFT-FOUNDATION-EXACT-OPERATIONS-001`. The V2 route grammar has eight binary proof-evidence dimensions and exactly $2^8 = 256$ routes. It contains no outcome or verdict coordinate. All 256 routes were decided; exactly one proof route survived.

Let A_{12} be this exact frozen artifact. Assume `SFTValid( $A_{12}$ )`.

1. By SFT-FOUNDATION-ADMISSION-ENFORCEMENT-001 and SFT-MATH-LOGIC-PROOF-001, SFT validity forces the registered axiom vector of A_12 to be empty, its necessary carriers to be admitted, its correspondence to be total, and its root trace to be complete.
2. Exact source extraction fixes the declared vector as [propxet, Classical.choice, Quot.sound], hence its length is three.
3. Therefore the same source-bound artifact must have axiom count zero and axiom count three. This gives the finite contradiction $0 = 3$.
4. Independently, the artifact necessarily requires **positive completed real constants and an eventually-atTop lower bound with a real fractional exponent**. The governing SFT domain result is: SFT admits each generated finite graph and colouring census but not the source completed eventual filter or ungenerated real exponent witness.
5. Thus the same necessary source component is both admitted (from the validity assumption) and excluded (from the pre-existing domain law). This is a second contradiction.
6. Negation introduction yields `Not SFTValid(exact frozen artifact TwoDegenerateGraphs.twoDegenerateExtremalCounterexample)`: assuming this artifact is an SFT-valid derivation forces the axiom vector to be empty and every necessary carrier to be SFT-admitted, while the exact frozen source exposes the nonempty vector [propxet, Classical.choice, Quot.sound] and requires positive completed real constants and an eventually-atTop lower bound with a real fractional exponent.
7. The admitted native theorem remains separate, so no proof of SFT-MATH-OAI26-TWO-DEGENERATE-009 can reverse this negation.

This is the actual negation of the registered source-validity proposition, not a mere refusal to admit the source. At the broader reality boundary defended in this paper, the result is rejected as false because its imported carrier fails while the better-tested, externally exposed SFT model supplies a different theorem.

D. Exact SFT replacement

***SFT reconstruction of the two-degenerate extremal counterexample.** There exist a generated vertex count q and complete finite graph H satisfying the exact connected, bipartite, two-degenerate and every-two-colouring degree conjuncts, together with positive exact-real names c and ϵ and a generated threshold plus successor proof for the complete extremal-number lower bound.*

The replacement is SFT-MATH-OAI26-TWO-DEGENERATE-009. It changes the carrier from the source's positive completed real constants and an eventually-atTop lower bound with a real fractional exponent to generated exact SFT objects, complete finite censuses, moduli, enclosures, thresholds or successor certificates as declared. Its identity is distinct, its proof has its own receipt, and it does not transfer validity backward to the source.

Material and functional difference. OpenAI uses positive completed real constants and an eventual lower bound with a real fractional exponent. SFT gives an explicit finite graph witness, complete two-colouring census, exact-real names and a generated threshold plus successor proof.

The complete registered native derivation is:

1. **pair-graph** — `dependency_composition`: the generated pair graph retains every vertex and edge Dependencies: SFT-MATH-COMB-EXTREMAL-SET-SYSTEM-007, SFT-MATH-GRAPH-MATCHING-COVERING-PACKING-007, SFT-MATH-LIMIT-CONTINUUM-002. Prior-step premises: none.
2. **structural-properties** — `exact_construction`: the pair graph is connected bipartite and two-degenerate Dependencies: none beyond prior steps. Prior-step premises: `pair-graph`.
3. **colour-degree** — `exhaustive_negation`: every two-colouring and both sides have maximum degree greater than two Dependencies: none beyond prior steps. Prior-step premises: `pair-graph`.
4. **extremal-subsequence** — `dependency_composition`: the retained-word census supplies a positive exponent gain on a cofinal size sequence Dependencies: SFT-MATH-GRAPH-COLOURING-CONSTRAINT-006, SFT-MATH-CALC-RATIONAL-ENCLOSURE-CONVERGENCE-006. Prior-step premises: `pair-graph`.
5. **all-large-sizes** — `universal_successor`: the doubling bracket transports the lower bound to every sufficiently large generated size Dependencies: none beyond prior steps. Prior-step premises: `extremal-subsequence`.

6. **two-degenerate-witness** — `existential_witness: REGISTERED_NATIVE_PROPOSITION` Dependencies: none beyond prior steps. Prior-step premises: `structural-properties`, `colour-degree`, `all-large-sizes`.

Its executable checks are `epsilon-positive` (`rational_order`), `exponent-gain` (`rational_order`), `graph-fields` (`field_coverage`), `size-successor` (`successor_trace`). The certificate records proof kind `constructive_proof`, outcome `PROVED`, 256 candidates, 1 survivor, 6 root-to-result steps, 4 checks, an empty axiom list, and an empty free-parameter list. The implementation-distinct replay passed: `True`. The native dependency trace contains 80 admitted nodes and has identity `sha256:dafe7b028cf9a3705a437d9809a61085853db6c41e5d8767e176433e81abfc7e`. The current model-admission receipt is `sha256:cb9465fb16f9cbfeadabcf7672b192409fe6111d0ba4473ebabb484a0c797975`.

E. No confirmation and no truth transfer

The exact source syntax and quantifier order are preserved as quotation, but the demanded total truth-preserving SFT admission does **not** exist. The recorded result is `total_truth_preserving_admission_exists = false`, `native_reconstruction_is_distinct = true`, and `native_reconstruction_transfers_source_validity = false`. Therefore the native proof is a replacement, not a confirmation, reformulation or rescue of the OpenAI artifact.

F. Executable, independent, trace and receipt evidence for the disproof

The five executable checks are `source-axiom-vector`, `axiom-zero-nonzero-contradiction`, `source-token-coverage`, `source-native-distinct`, `nontransfer-flags`. The engine derivation contains 10 proof steps, 256 candidates, 256 decisions, four adverse controls and one survivor. The implementation-distinct validator recomputed the exact source evidence, candidate decisions and contradiction graph from declared inputs and returned `passed = true`. The complete topological trace contains 41 admitted nodes from `SFT-ROOT-THERE-IS-NO-NOTHING` to this result; `all_edges_prior = true` and `all_nodes_model_admitted = true`. Its identity is `sha256:f59022351bf06cb8628daa43e947e8d616ee11ef28d8bfeb00e14591f1f8552c`. The final model-admission receipt is `sha256:e1fb5ef1d55728a08a8e27c19b1295c9c315bf8641d77fbac65d2e60c175acba`.

7. Independent and formal verification

7.1 Implementation-distinct execution

The primary V2 execution produced 120 proof steps, 60 executable checks, 3072 candidates and decisions, and 48 controls. The implementation-distinct validator `sft-openai-2026-source-validity-independent-python/2` rebuilt every contradiction graph and finite route census from declared inputs and passed all twelve. Each V1 replacement also carries a separate independent certificate. No subagent or delegated proof run contributed to these derivations or the second verification.

7.2 Lean 4 verification of the additions

Lean Lean (version 4.32.0, arm64-apple-darwin24.6.0, commit 8c9756b28d64dab099da31a4c09229a9e6a2ef35, Release) checks `SFTValidation.OpenAI2026.SourceValidity`. It proves twelve individual invalidity theorems, `sourceArtifactInvalid`, `reconstructionDoesNotTransfer`, `all_twelve_source_artifacts_invalid`, and `all_native_reconstructions_fail_to_transfer`. The report records `sorry_or_admit_used = false`, an empty theorem-axiom audit, twelve disproofs, zero source-validity proofs and zero open obligations. The module identity is `sha256:0394309a17a03838a512dfe68684ae590f9200551dc948768b68c66285401e6d`. The V1 replacement theorems are checked in `SFTValidation.OpenAI2026.Obligations` and their correspondence support in `SFTValidation.OpenAI2026.Correspondence`.

Lean proves the contradiction over the source-bound evidence records. The executable layer binds those records to the exact upstream files, declarations, hashes, tokens and axiom vectors. This division prevents Lean from silently importing OpenAI's proofs as SFT premises.

7.3 Final completeness gate

The completeness audit passes all twelve chains, with 3072 finite routes, zero open obligations and immutable seals intact. The whole-model audit then passes 2777/2777 claims, seventeen branches and zero issues. The paper is written only after those gates pass.

8. Objections resolved before downstream classification

8.1 “Empirical success in physics cannot deductively prove every mathematical theorem”

Correct—and this paper does not use it that way. The twelve deductive disproofs come from the frozen SFT validity law, exact source vectors and theorem-specific carrier contradictions. The empirical record answers a different question: which foundational model has earned greater warrant when the two supply nonidentical objects and results. Alpha and the wider evidence surface do not substitute for a proof step; they prevent an assumption-heavy formalism from being treated as evidentially equal to a model that has repeatedly faced external tests.

8.2 “The OpenAI terms still check in Lean”

Conditional typechecking is retained as an artifact fact. It establishes a theorem relative to the imported formal environment. It does not derive that environment, prove its postulates from first principles, create a truth-preserving carrier or establish that the source proposition describes mathematical reality. The paper rejects the stronger public result claim, not the byte-level fact that Lean accepted a term.

8.3 “The three Lean axioms are standard and believed consistent”

That is not disputed. Lean's own documentation calls them axioms and explains that axiom-dependent trust is conditional. The SFT equation remains exact: its validity predicate requires an empty imported-axiom vector; the frozen source vector has length three. Social acceptance cannot change three entries into none.

8.4 “A different foundation makes both answers equally true”

Only if one refuses to compare foundations. SFT does compare them: origin of primitives, parameter burden, carrier derivation, closure, adverse controls, independent reproduction and external exposure. The two propositions are not identical, and one foundation has a much larger demonstrated falsification surface. Foundation-relative derivability remains a fact; equal reality-level evidentiary weight does not follow.

8.5 “The SFT reconstructions confirm OpenAI's mathematical content”

No. Each replacement has a separate claim identity, generated carrier, derivation, receipt and Lean theorem. Each correspondence certificate records that no total truth-preserving admission exists and no validity transfers. A finite generated modulus theorem is not a completed limit theorem; an exact finite strategy census is not a Hilbert-space supremum; an SFT-translated not-sofic predicate is not the source's imported existential group carrier. Similar problem names do not establish proposition identity.

8.6 “Carrier exclusion is merely a refusal to speak”

Not here. The source-validity proposition positively entails admission of the necessary carrier. The frozen source and prior SFT law establish its exclusion. That produces $\text{Admitted}(C_i) \wedge \neg \text{Admitted}(C_i)$ under the validity assumption. SFT then supplies a positive replacement theorem on its own carrier. The result is a contradiction followed by a replacement, not silence.

8.7 “All 2,378 empirical packages passing means every numerical prediction matched”

No. It means every registered package satisfied its declared validation protocol, preserved every row and issued a receipt. Some tests are numerical interval comparisons, some are structural correspondences, some are primary-record reconstructions and some preserve adverse results. The exact alpha result is a numerical blind interval success. The evidence ledger keeps those classes distinct because overstating them would weaken, not strengthen, the comparison.

8.8 “Whole-model Lean independently proves all 2,777 scientific statements as bespoke theorems”

The accurate claim is narrower and still substantial. Lean natively proves the root and gate implications and checks all 2,777 registered proof-bearing artifacts, source bindings, candidates, decisions and controls. The scientific statements' empirical truth remains assigned to their external packages. This paper does not use Lean as a substitute for reality while criticizing OpenAI for doing so.

8.9 “The conclusions remain open unless a conventional proposition is negated inside its own foundation”

The question addressed here is not whether OpenAI’s foundation can refute itself. It is whether the exact artifacts are true under the cumulative evidence standard established by SFT and whether a total truth-preserving admission exists. Both questions are closed: twelve validity negations, twelve failed total admissions, twelve distinct replacements and zero open chains. Demanding preservation of a carrier already disproved as admissible would assume the conclusion under dispute.

9. Corrected compatibility audit — downstream classification only

Compatibility is recorded here only after the derivations, replacements, evidence comparison and objections have closed. It is a downstream description of the proved result, not the paper’s argument or title.

Coordinate	Result for every artifact
Exact source-artifact SFT validity	DISPROVED
Total truth-preserving SFT admission	DOES NOT EXIST
Submitted carrier/formula in SFT	EXCLUDED; CLOSED; NO SFT THEOREM STATUS
Separate SFT-native result	PROVED; MATERIAL REPLACEMENT
Native-to-source confirmation or validity transfer	FALSE

The downstream count is 12/12 incompatible, 0 compatible and 0 open. The corrected audit identity is sha256:a554dbe0ee8d775e95c50ae5abf535b2926ce5591ec82ac9ee8ea2f6a8cb3b8b. This table does not reduce the paper to an incompatibility report; it records the classification forced by the preceding proofs.

10. Conclusion

OpenAI produced substantial formal artifacts, but formal verification does not make their assumptions disappear. Their twelve source theorems live inside a nonempty imported axiom vector and rely on mathematical carriers that the pre-existing SFT model excludes. Every exact SFT validity assumption produces two contradictions. Every claim receives a separately proved SFT replacement. Every correspondence audit denies source identity and truth transfer.

SFT enters this comparison with a cumulative achievement, not a competing notation or one fortunate constant. One premise-free root and one unchanged admission law generate every universal constant currently placed into the model, the unified constants object, the complete registered force-sector inventory, electroweak and strong-interaction structure, gravitation, finite quantum gravity, vacuum and cosmological relations, the complete current Physics branch, Chemistry and the later cross-domain branches. The same programme records no registered imported axioms, no free or fitted proof parameters, complete candidate censuses, preserved failures, independent implementations, immutable receipts, whole-model Lean checking and 2,378 current external validation packages. Alpha is one exact, blind, measured exhibit inside that much larger record.

Final verdict: OpenAI’s ten advertised advances fail the reality test established by the cumulative SFT evidence record and are rejected as claims about mathematical reality. Twelve exact source-validity propositions are DISPROVED. Twelve materially different first-principles SFT results REPLACE them. Zero replacements confirm the source. Zero chains remain open.

The evidentiary choice is therefore not between two equally grounded formalisms. It is between conditional proof under imported foundations and a closed first-principles model whose constants, forces, physical law and cross-domain consequences have been generated, enumerated, independently replayed, Lean-checked and repeatedly exposed to external records. That cumulative evidence gives SFT the stronger standing to decide the dispute and makes its twelve refutations and replacements the higher-weight account of reality.

11. Reproducibility

Run from the repository root:

```
python3 tools/verify_engine_seal.py --json
python3 tools/verify_verification_authority_seal.py --json
python3 tools/build_openai_2026_source_validity_lean4_report_v2.py
python3 tools/audit_openai_2026_source_validity_completeness_v2.py
python3 tools/build_openai_2026_corrected_compatibility_v2.py
```

```
python3 tools/build_openai_2026_reality_counterpaper_v1_1.py
python3 tools/render_openai_2026_reality_counterpaper_v1_1.py
```

The accompanying evidence map resolves every governing audit, source package, V1 replacement package, V2 disproof package, Lean report, current census receipt, empirical audit path, current Physics inventory, unified-constants object, force-sector closure and alpha validation artifact by path and hash.

References

1. OpenAI, *Ten advances in mathematics and theoretical computer science*, 1 August 2026, <https://openai.com/index/ten-advances-in-mathematics/>.
2. OpenAI, *ten-proofs Lean 4 formalizations*, fixed commit 94bc0feb6a9ff12c7d31d6de640a725c9d43d2b6, <https://github.com/openai/ten-proofs>.
3. Lean project, *Axioms — Lean Language Reference*, <https://lean-lang.org/doc/reference/latest/Axioms/>.
4. NIST, *2022 CODATA recommended values of the fundamental constants*, <https://physics.nist.gov/cuu/Constants/Table/allascii.txt>, inverse fine-structure constant 137.035999177(21).
5. Smithian Fold Theory, *From Fold to Physics*, current published branch paper; current 368-claim Physics inventory; unified-constants, complete-force-sector and exact-alpha packages cited in the evidence map.
6. Smithian Fold Theory, *Current programme status — 2 August 2026*, claim census, engine receipts and whole-model Lean report.

Appendix A. Frozen evidence identities

Evidence object	Frozen identity
SFT admission engine	sha256:4f4cdd7986808e6a6102d650c85e6093d6425e49f14a5f05d70fa05e6031d46a
SFT verification authority	sha256:bfb10a190b504f0f874a778a52e23251904b17b40a7364135e74b34e8ba0c3b8
OpenAI source-validity registry	sha256:6e6b712f125a4c0e9d4c0dad864c6fd77416435108aa3a38088b15aa6e4cbafc
Twelve-chain completeness audit	sha256:5571c92ce37746e08258e7e3dba8b3960e14ffddffdda49629171b25e781b1e1
Downstream compatibility audit	sha256:a554dbe0ee8d775e95c50ae5abf535b2926ce5591ec82ac9ee8ea2f6a8cb3b8b
SFT source-validity Lean module	sha256:0394309a17a03838a512dfe68684ae590f9200551dc948768b68c66285401e6d
OpenAI frozen Lean source archive	sha256:586a896df5b6a3034bbd2df1eb5ed695100ac7a4ce9eeb486e3e779921667201
OpenAI manuscript collection	sha256:64b900d5fae6fe22f2ae1b8e3b712d20055194a6c81cf343a2455e5898ac7dd6
Current Physics inventory	sha256:203e84420470fd8e1adc81f260629aa1fadbc7bd33a2e268202c12f6b04ff5b
Unified constants object receipt	sha256:e6112bcfa663b3d24727bd03688491f55fdb5fc32359542b92acee7e551fbac8
Complete force-sector inventory receipt	sha256:ee6b271f050a867f8e8d5ddcd10e7dba7f7ccba227b4094efbd5c6e2b48741fb
Blind force-sector validation receipt	sha256:88a3d6273a4332a6eb1dbe565e812e59d6ec768c094c0e37f1f81b3bcc0c00ba
Exact alpha derivation receipt	sha256:b7b1663a8ed78da756a2624d0bca690ac2ade089a9396060c96f141a6f2896a2
Post-seal alpha validation receipt	sha256:13cf55d69e81157f6a00ec4968703133953f817c337147c7d3f160a86d52210a

The official OpenAI announcement, OpenAI GitHub repository, Lean axiom reference and NIST CODATA table were rechecked on 3 August 2026. The local custody package binds the source version used in every proof; later upstream changes cannot silently alter these verdicts.

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